פקולטה להנדסת תוכנה

בדיקת מערכות ספרתיות ואימות תכנונן

פרק 6

LVS, PEX

cādence™

Assura Physical Verification

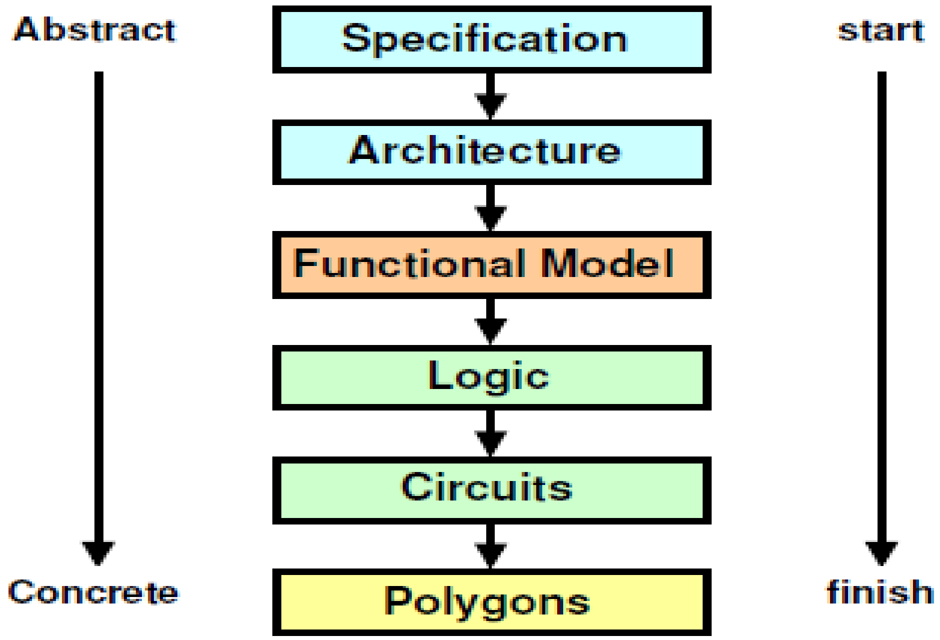
SYNOPSYS®

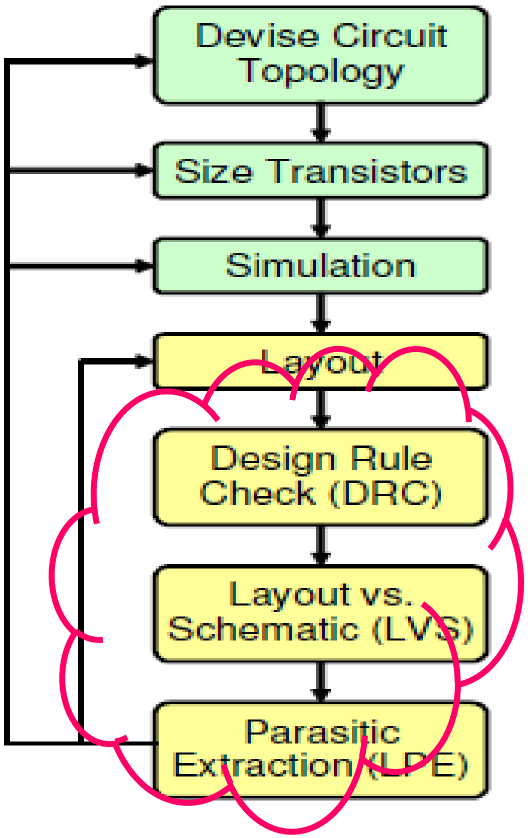
Hercules™

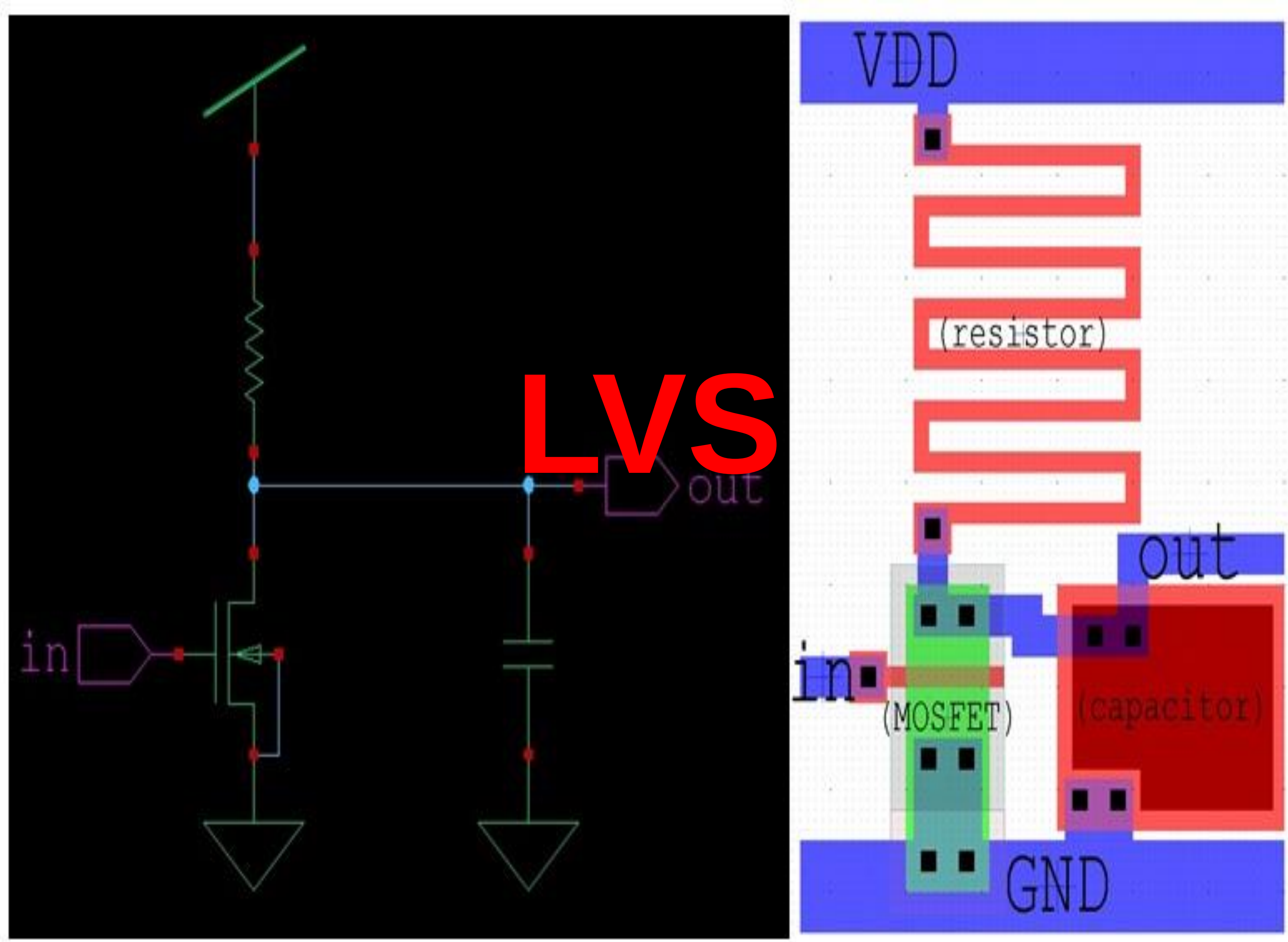
Mentor Graphics®

Calibre

מתודולוגית תכנון

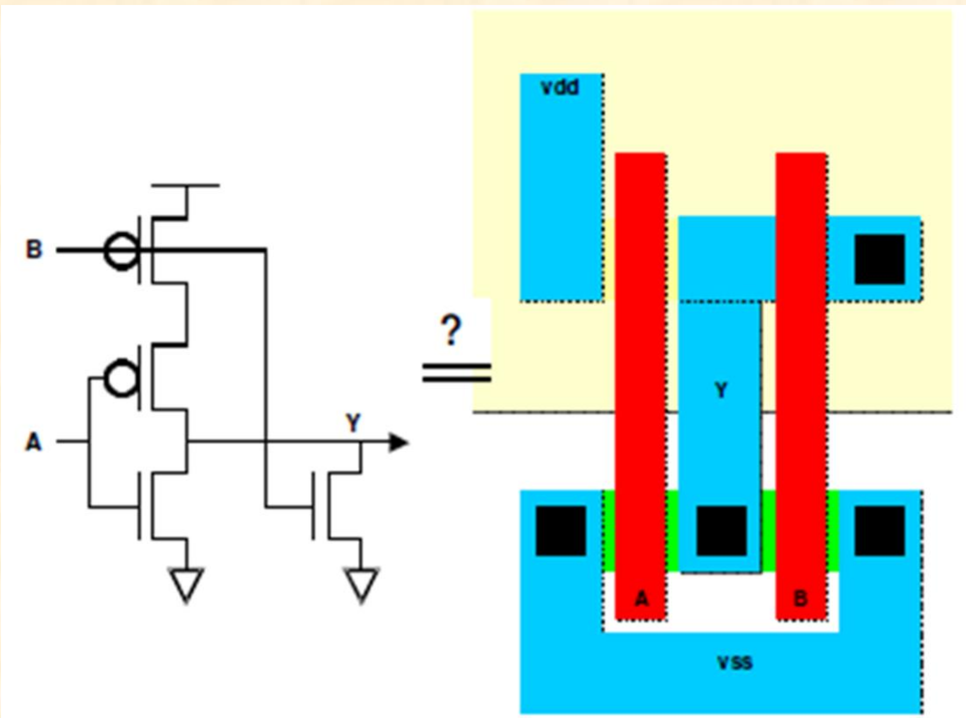





LVS
in
out
VDD
(resistor)
out
in
(MOSFET)
(capacitor)
GND

Layout Versus Schematic (LVS)

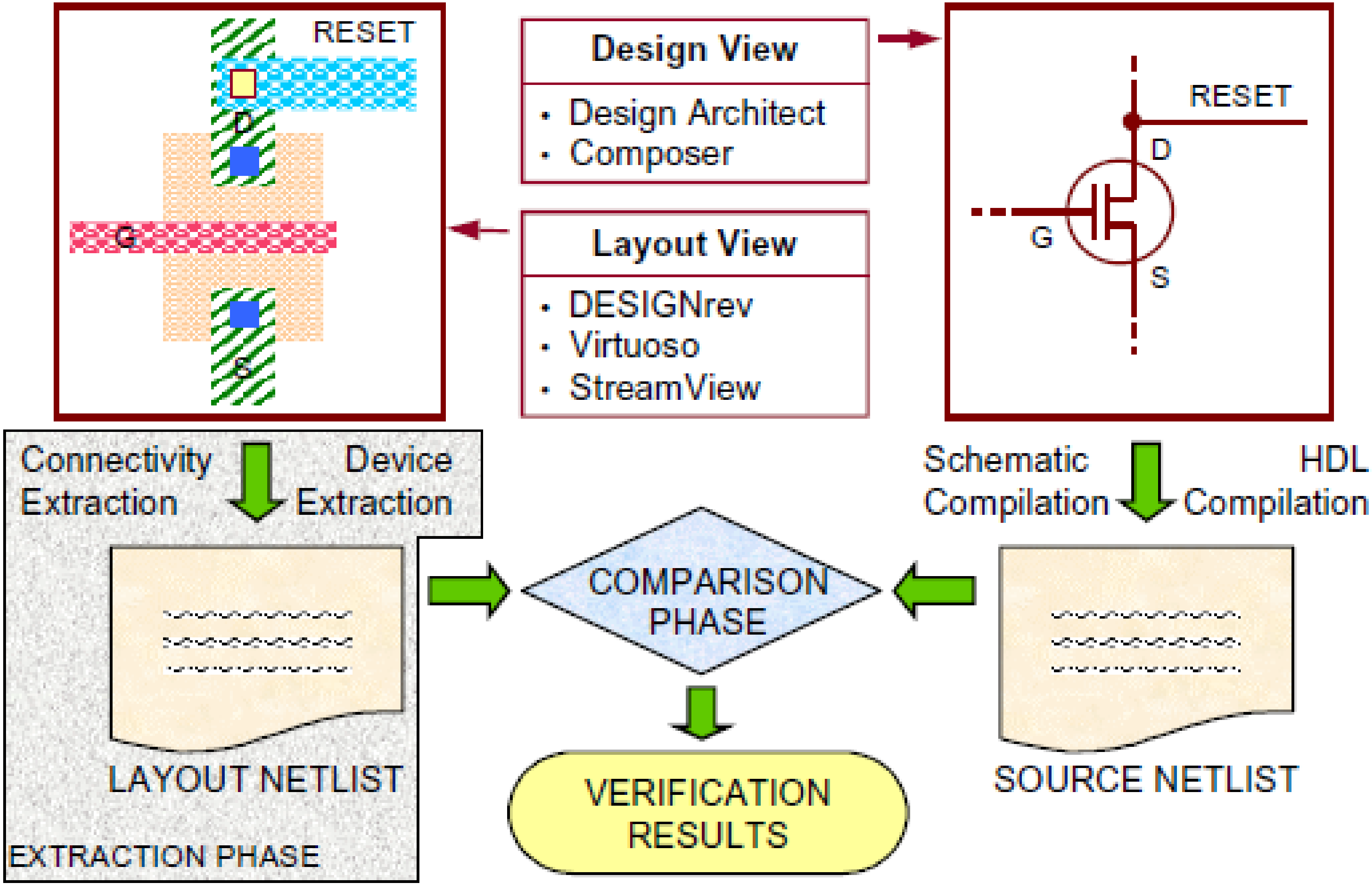
- <u>המטרה</u> השוואה בין netlists : סכמטי ו-layout
- השווה טרנזיסטורים וחיבורים (להתעלם parasitics)
- הנפקת שגיאה אם שני netlists לא שווים
- חשוב לתכנונים גדולים



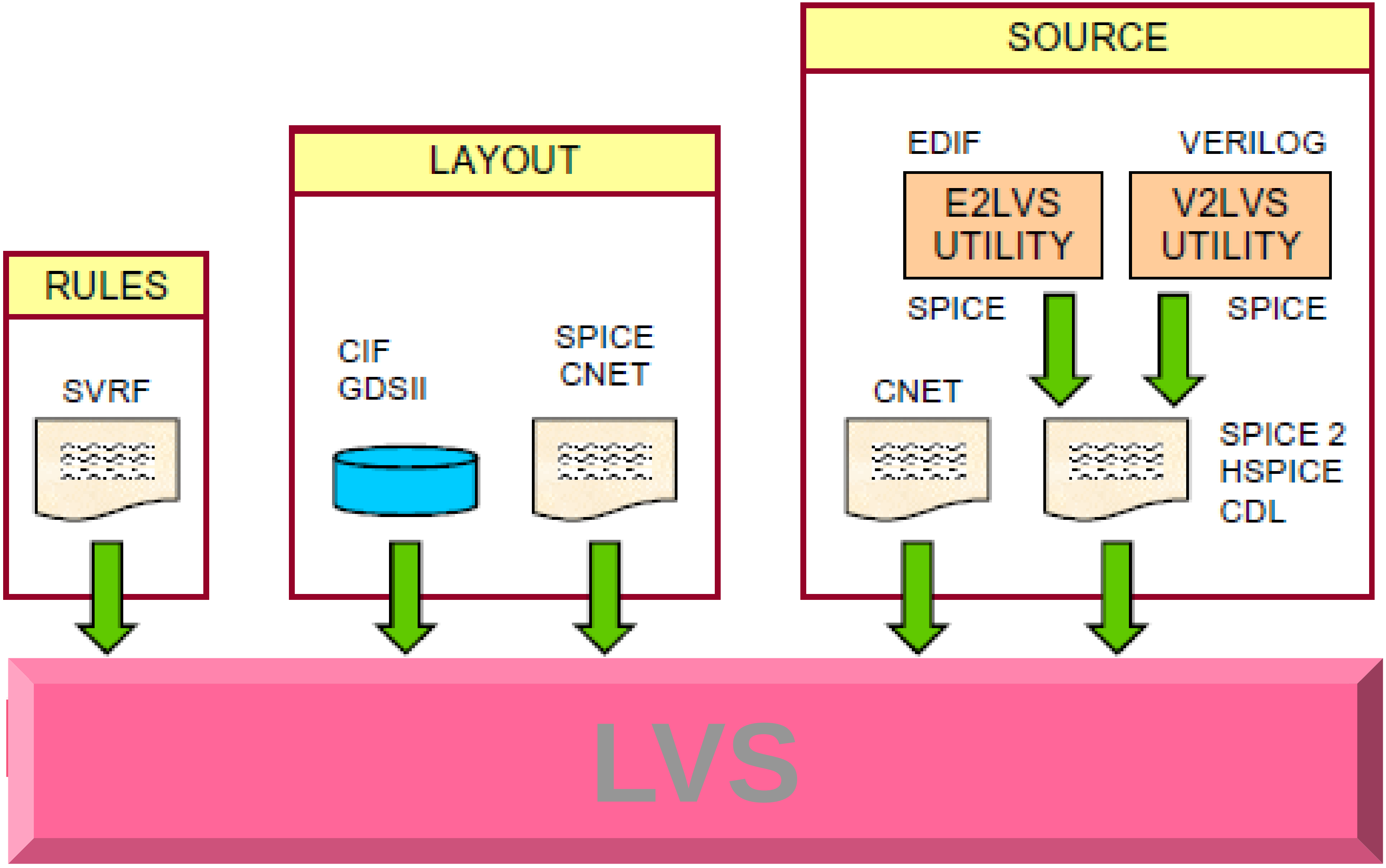
Layout Versus Schematic (LVS)

- LVS מחלץ netlist מ-layout על ידי ניתוח פוליגוני חפיפות
- LVS משווה את ה-netlist המחולץ עם ה- netlist הסכמטי המקורי
- כאשר סתירות התרחשו, מנסים מצוא את המקום

Layout Versus Schematic (LVS)

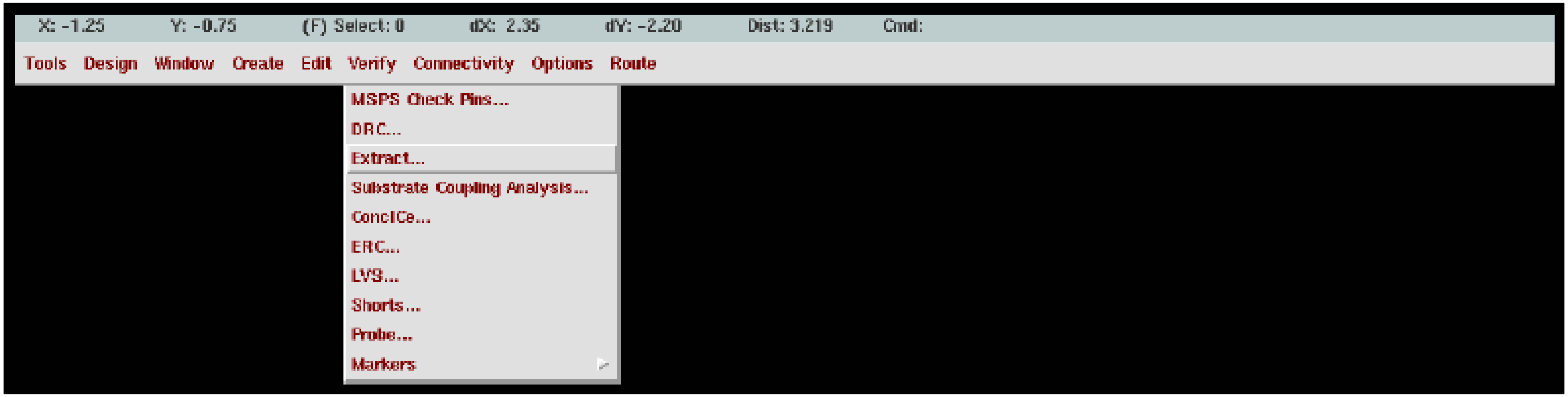


LVS runset

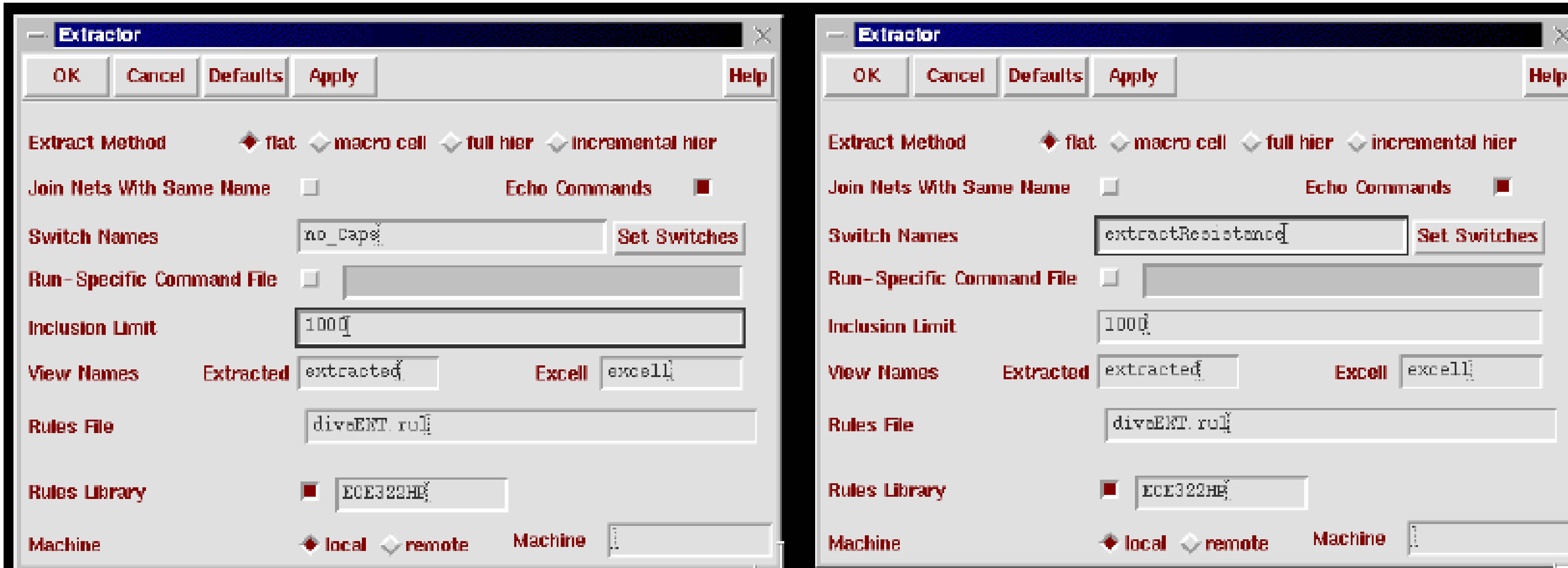


RULES
SVRF
LAYOUT
CIF
GDSII
SPICE
CNET
SOURCE
EDIF
VERILOG
E2LVS
UTILITY
V2LVS
UTILITY
SPICE
SPICE
CNET
SPICE 2
HSPICE
CDL
LVS

Extractions – עקירות

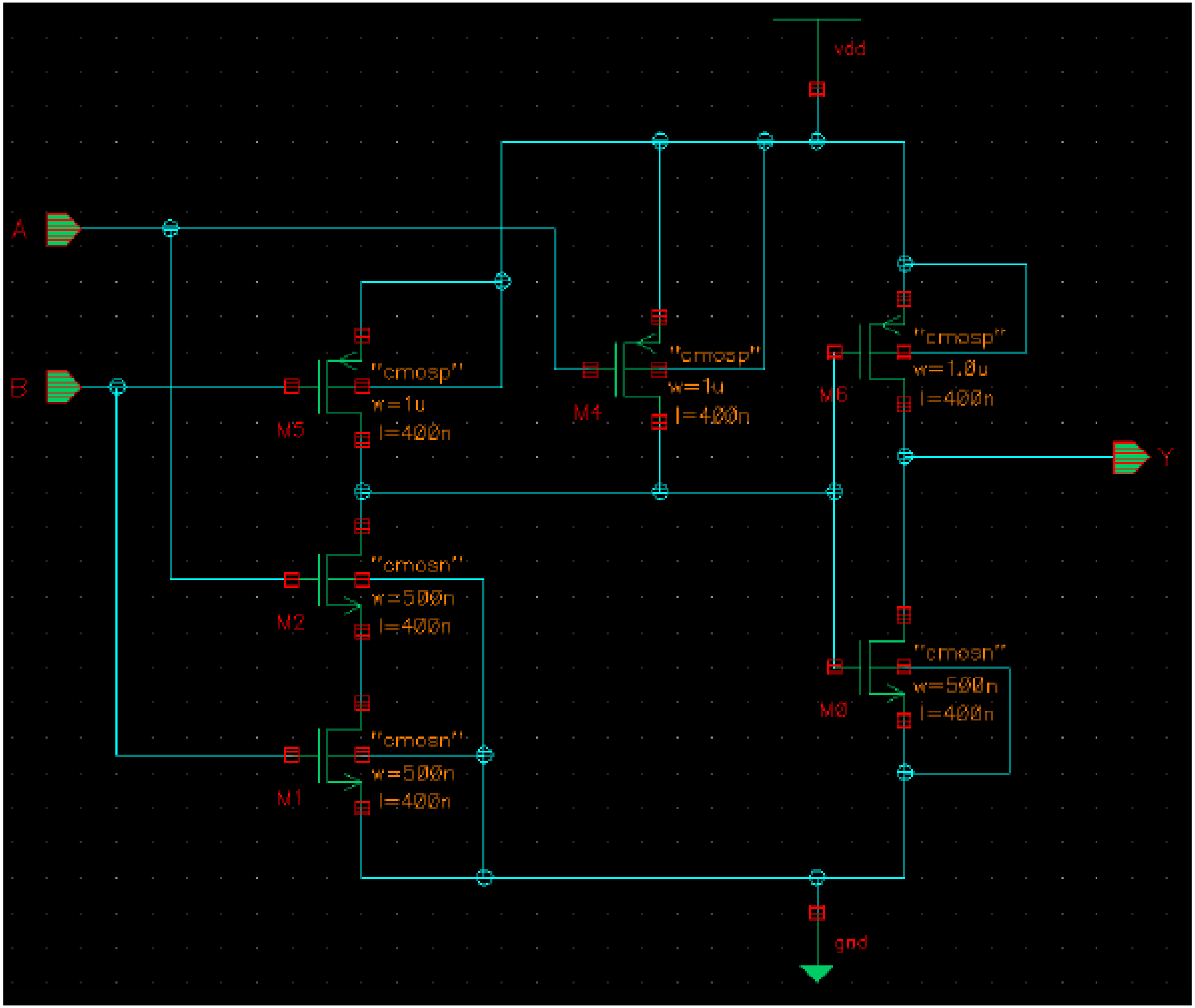


אפשרויות עקירות



LVS Options SPICE Options

דוגמת LVS

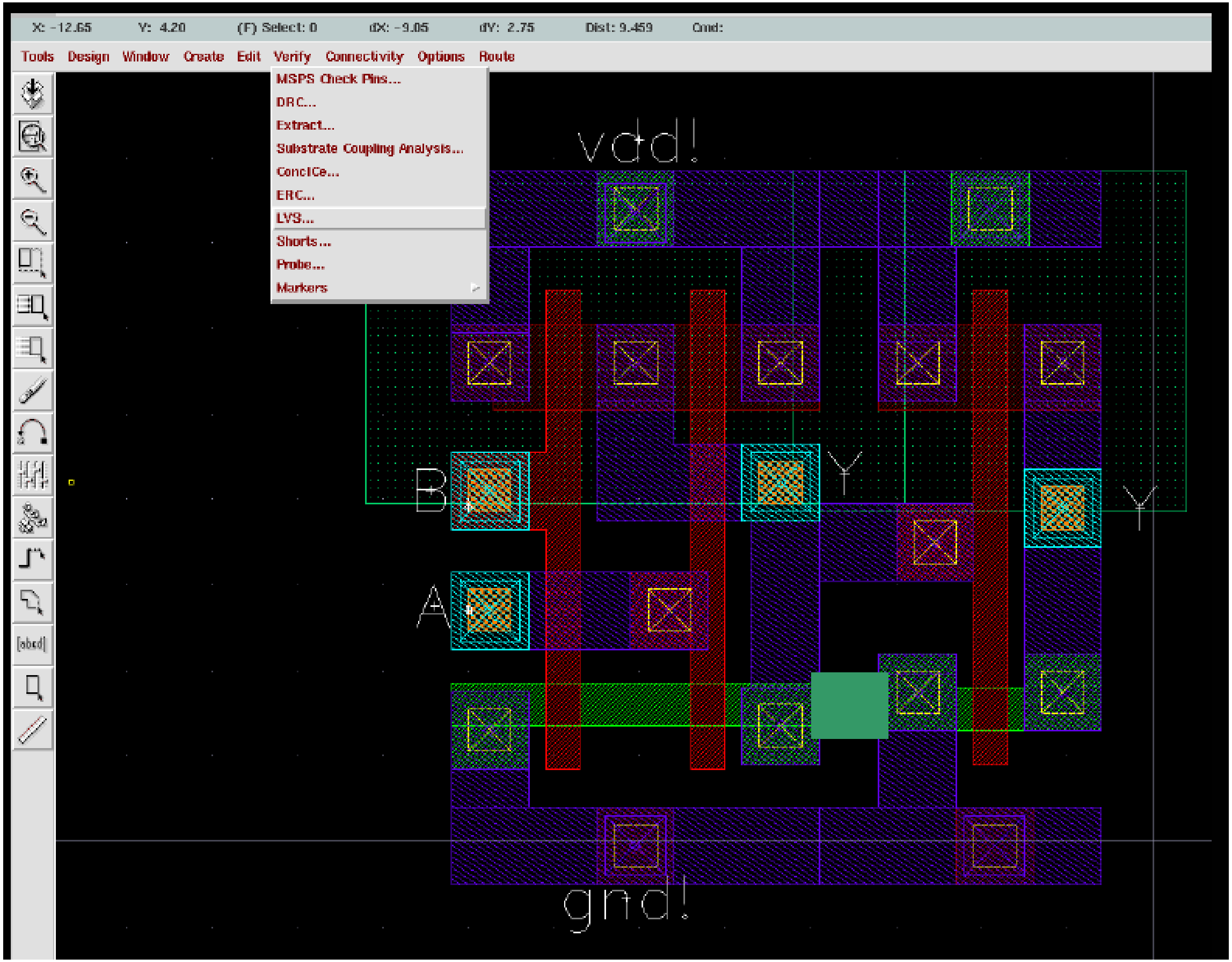


vdd
A
B
"cmosp"
w=1u
l=400n
M5
M4
"cmosp"
w=1.0u
l=400n
M6
Y
"cmosn"
w=500n
l=400n
M2
M1
M0
gnd

דוגמת LVS

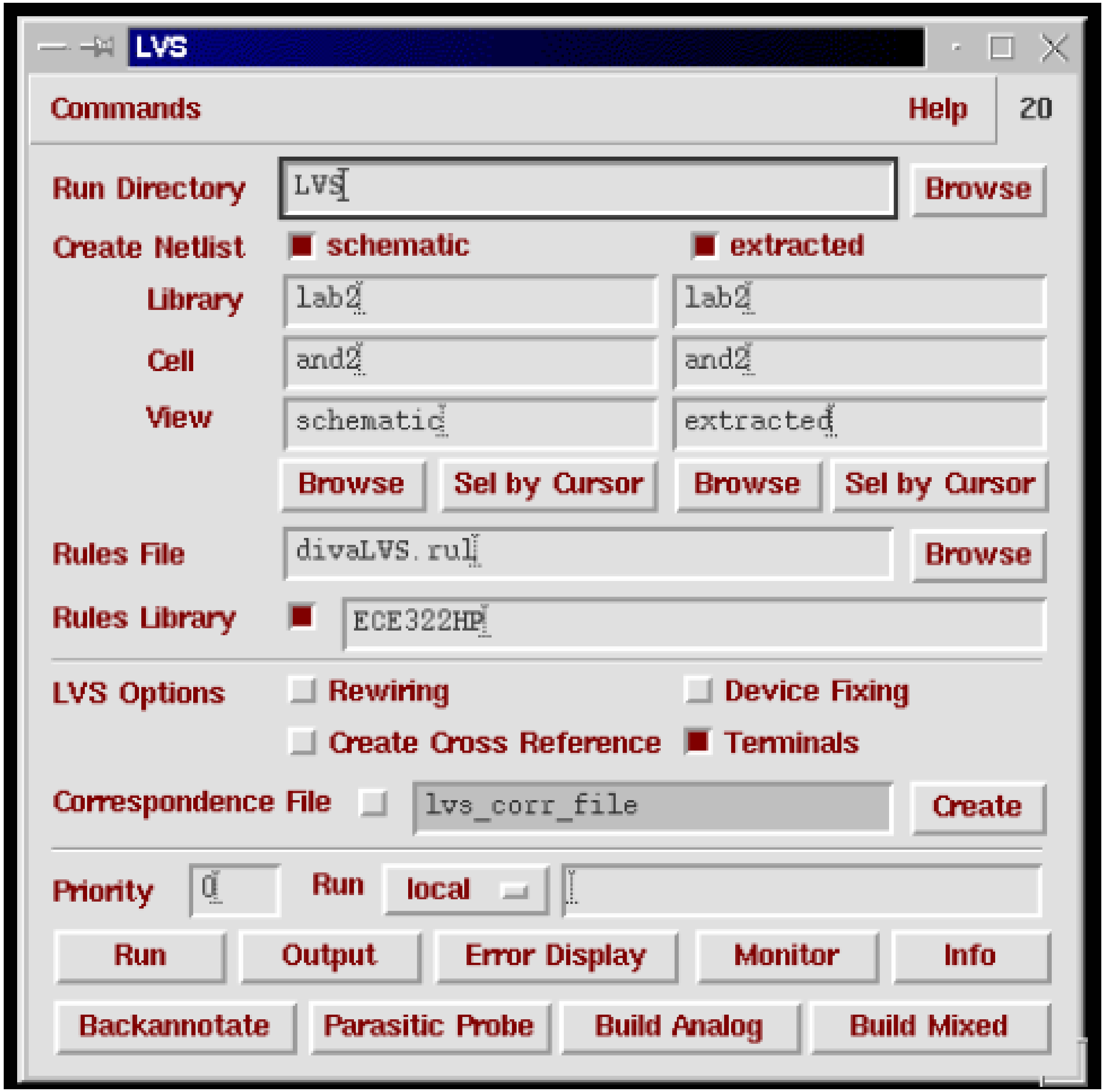

X: -12.65 Y: 4.20 (F) Select: 0 dX: -9.05 dY: 2.75 Dist: 9.459 Cmd:
Tools Design Window Create Edit Verify Connectivity Options Route
vdd!
B
A
Y
Y
gnd!

דוגמת LVS



X: -12.65 Y: 4.20 (F) Select: 0 dX: -9.05 dY: 2.75 Dist: 9.459 Cmd:
Tools Design Window Create Edit Verify Connectivity Options Route
MSPS Check Pins...
DRC...
Extract...
Substrate Coupling Analysis...
ConclCe...
ERC...
LVS...
Shorts...
Probe...
Markers
vdd!
B
A
Y
Y
gnd!

דוגמת LVS


LVS
Commands
Help
20
Run Directory
LVS
Browse
Create Netlist
schematic
extracted
Library
lab2
lab2
Cell
and2
and2
View
schematic
extracted
Browse
Sel by Cursor
Browse
Sel by Cursor
Rules File
divaLVS.rul
Browse
Rules Library
ECE322HP
LVS Options
Rewiring
Device Fixing
Create Cross Reference
Terminals
Correspondence File
lvs_corr_file
Create
Priority
0
Run
local
Run
Output
Error Display
Monitor
Info
Backannotate
Parasitic Probe
Build Analog
Build Mixed

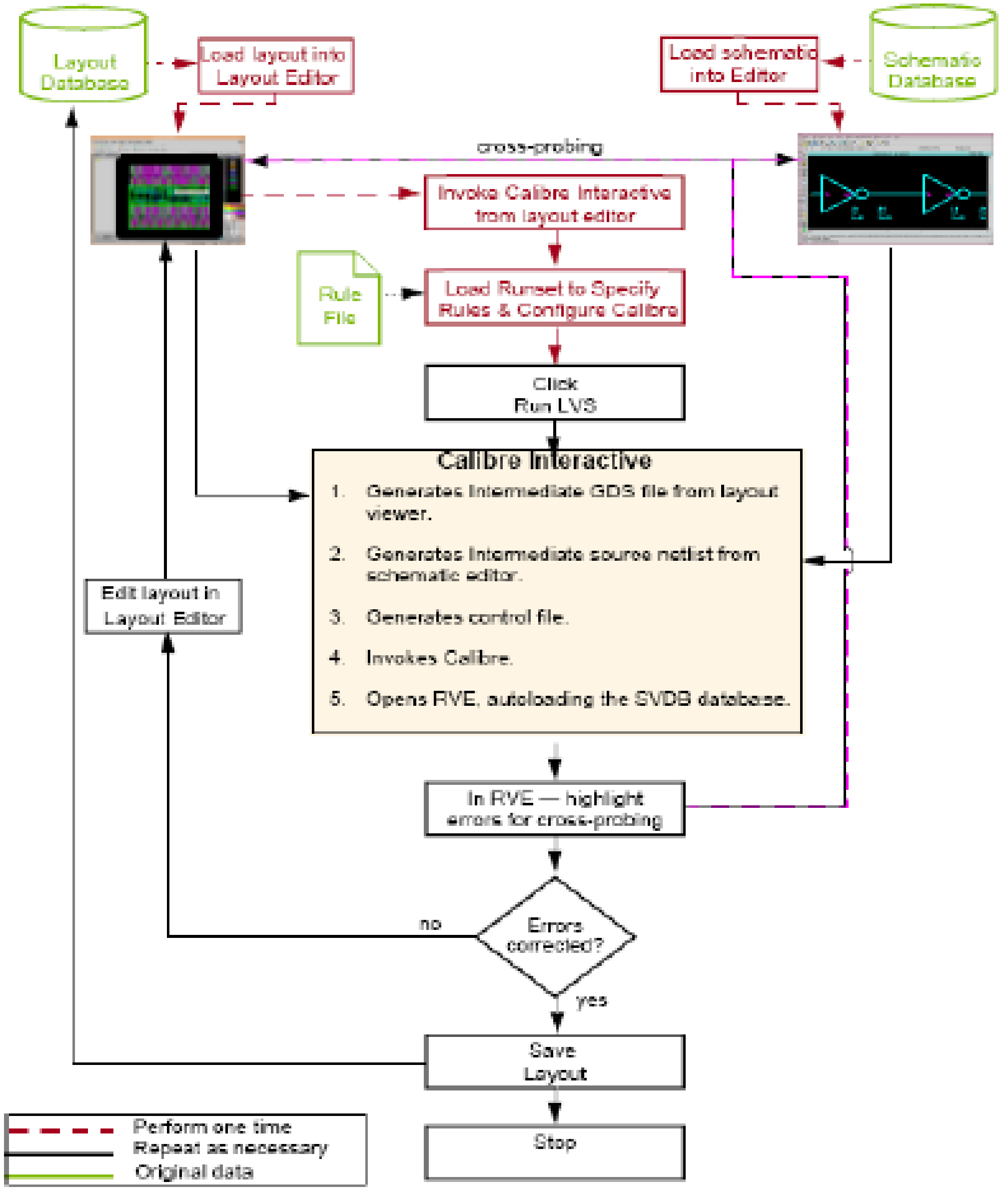
לא תואם!!!

```
Like matching is enabled.
Using terminal names as correspondence points.
Net-list summary for extracted view
        count
     6          nets
     3          terminals
     3          pmos
     3          nmos
Net-list summary for schematic
        count
     7          nets
     5          terminals
     3          pmos
     3          nmos
```

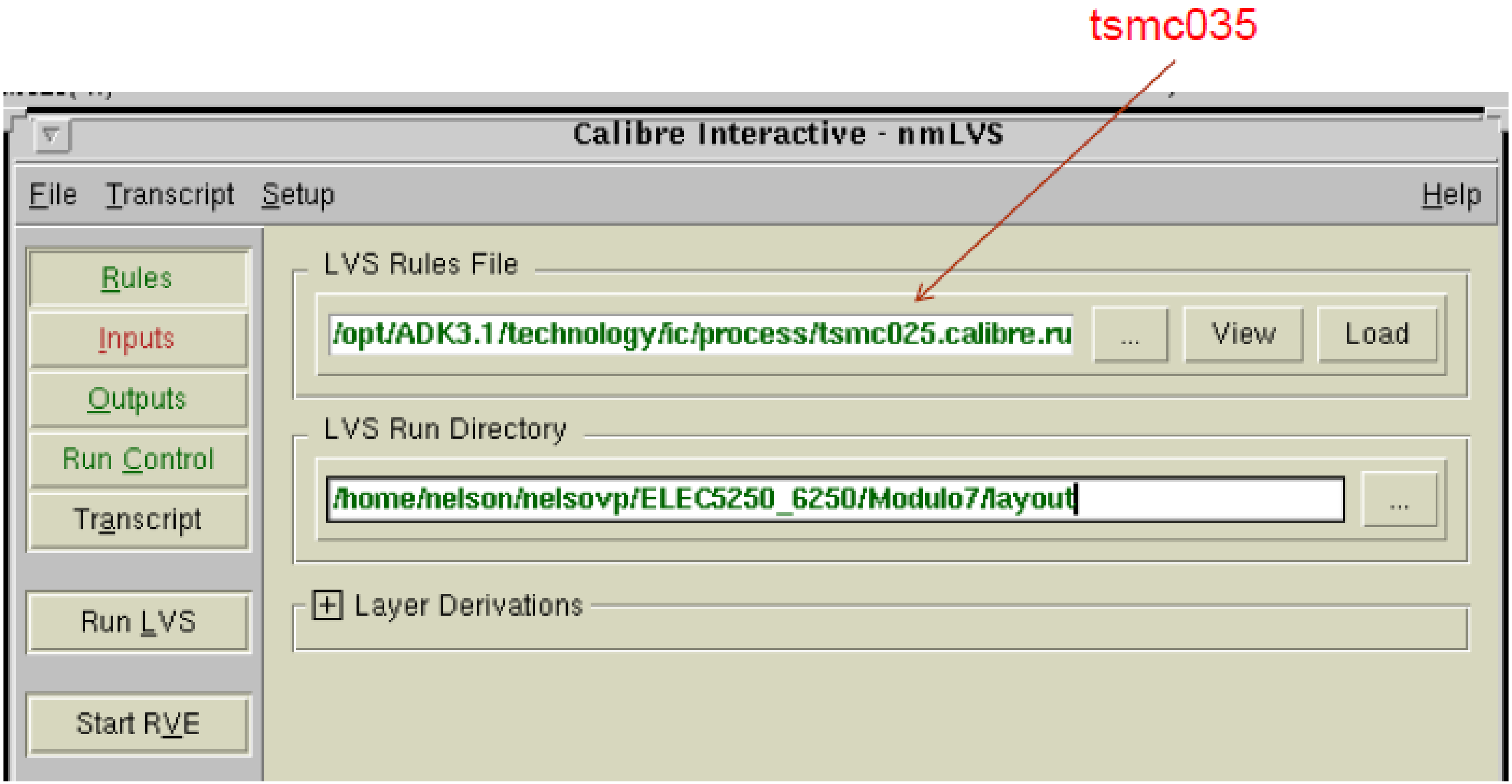
לא תואם!!!

```
                 layout  schematic
                  instances
un-matched        2     2
rewired                 0     0
size errors             0     0
pruned                  0     0
active                  6     6
total             6     6

                    nets
un-matched        2     3
merged                  0     0
pruned                  0     0
active                  6     7
total             6     7

                  terminals
```

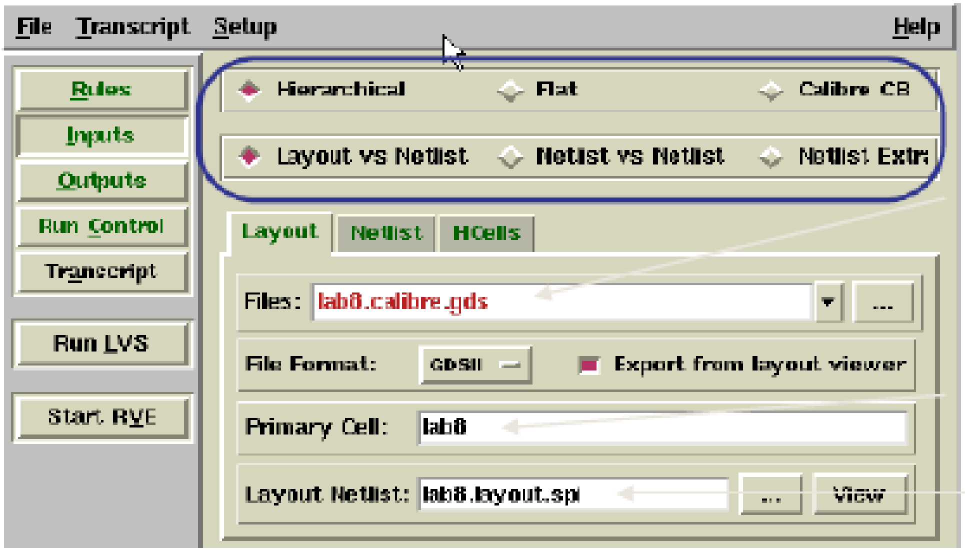
תהליך בדיקות LVS



טעינה של קובץ חוקי LVS



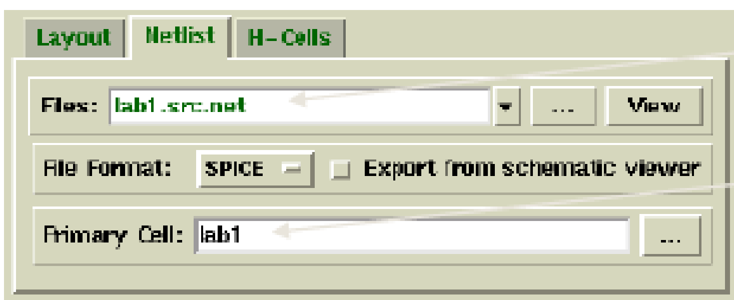
קלט לריצת LVS



Layout to be extracted by Calibre (GDSII format)

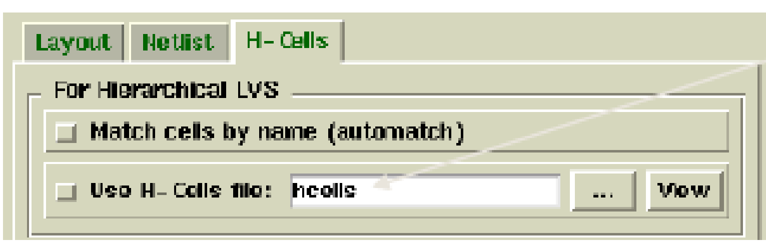
Layout top cell name

Extracted layout netlist



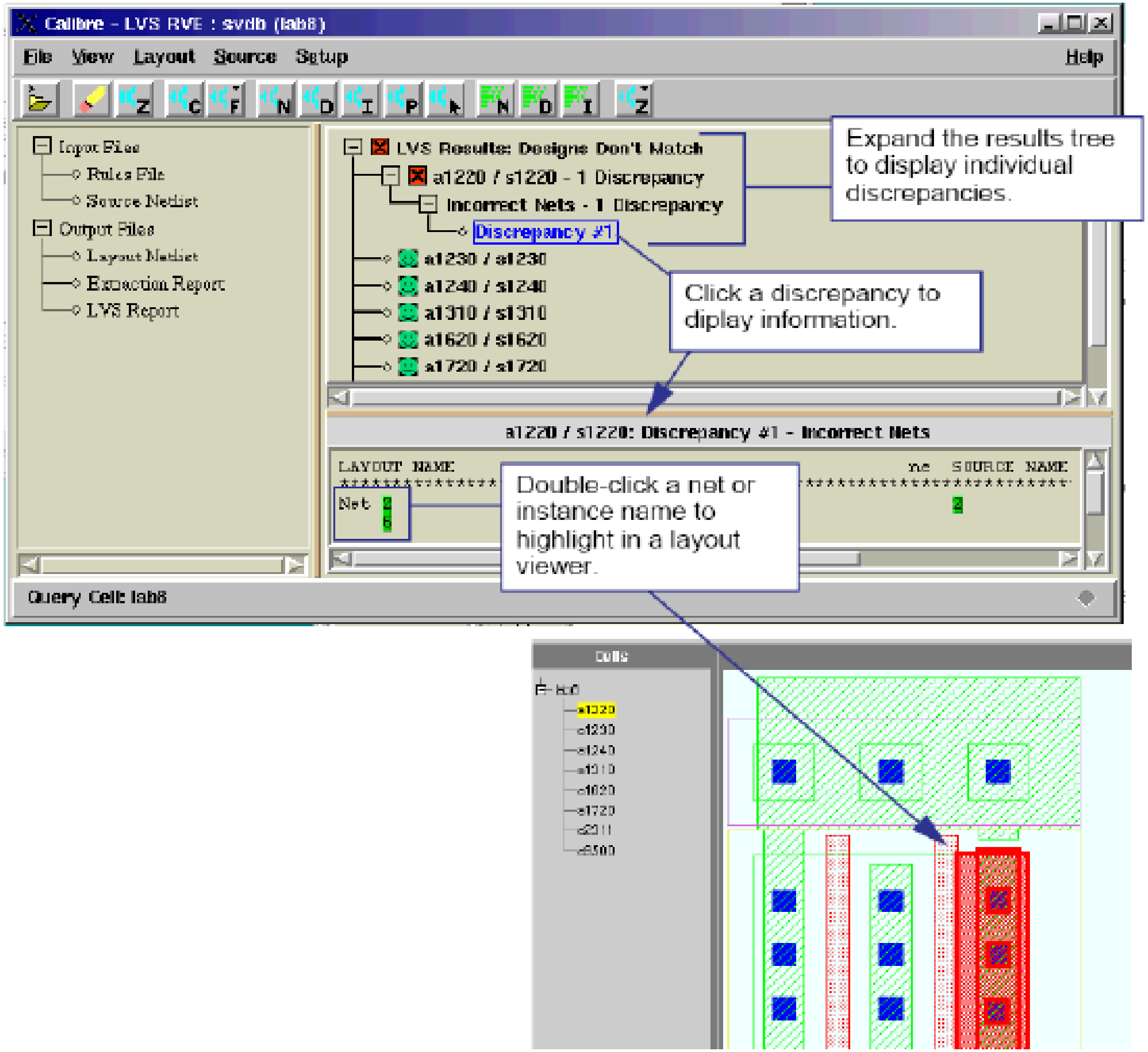
Source netlist created in DA-IC

Schematic name



Hierarchical cells file:
$ADK/technology/adk.hcell

חוקרים את התוצאות של LVS



קובץ חיקי LVS: תחביר

- **Layer Definition**

  Example:

  LAYER metal1 10 // **LAYER *name original_layer***

- **Layer Operation (DRC Check)**

  Example:

  INTERNAL oxide > 12 < 14

- **Specification Statement**

  Example:

  LVS REPORT name

- **Connectivity Definition/Extraction**

  Example:

  CONNECT layer1 layer2

קובץ חיקי LVS: תחביר

- Device Recognition

  Example: Resistor named “res”

  DEVICE R res metal1 (pos) metal2 (neg) [1.1]

  // Resistor with resistivity defined at 1.1 Ohms/square

- Device Statements
  - device mn ngate ipoly nsd nsd s_pwell [0]
  - device r(pl) rpoly ipoly ipoly [20000]
- Gate Recognition Statements
  - LVS GROUND NAME VSS VSS1
  - LVS POWER NAME VDD VCC
  - LVS RECOGNIZE GATES ALL

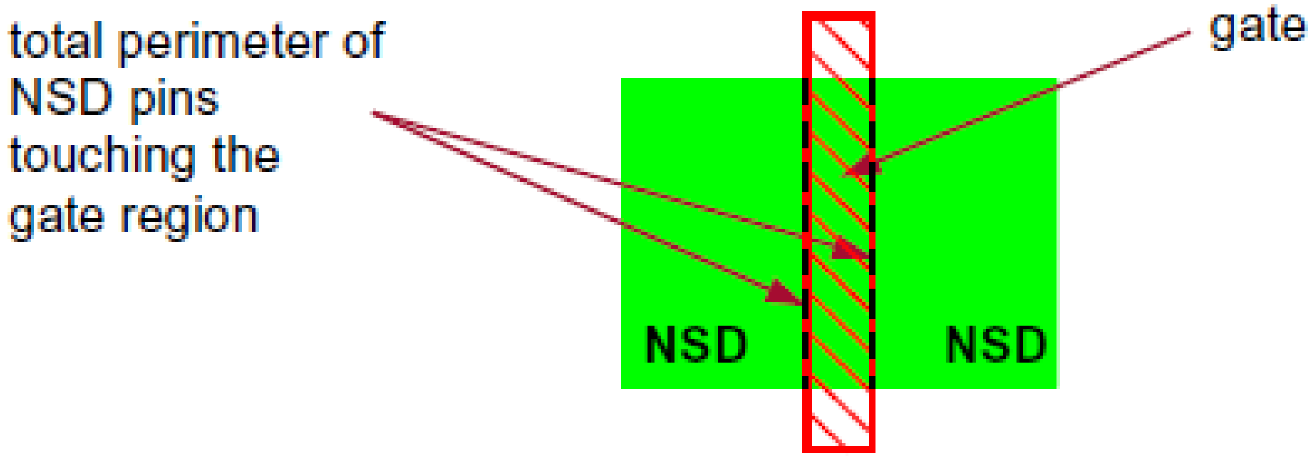
רעיונות יסוד של זיהוי רכיבים

- זהה מופעים של רכיבים מן הגיאומטרית layout
- חשב את המאפיינים של המופעים של הרכיבים

<u>הגדרות מובנות של רכיבים:</u>

- **MOS Transistor (M MN MP MD ME)**
- **Diode (D)**
- **Capacitor (C)**
- **Resistor (R)**
- **Bipolar Transistor (Q)**

דוגמת הרכיב: MOS



DEV MN (NMOS) GATE GATE(G) NSD(S) NSD(D) PWELL(B) [0.5]

//N-TYPE TRANSISTOR OF MODEL NMOS

//weffect PROPERTY SPECIFICATION OF 0.5 FOR BENT GATES

- Length and width are properties computed for MOS transistors by default (in meters)
- Width, “W”, is taken as half the total perimeter of NSD pins touching the gate region
- Length, “L”, is calculated as gate area divided by width

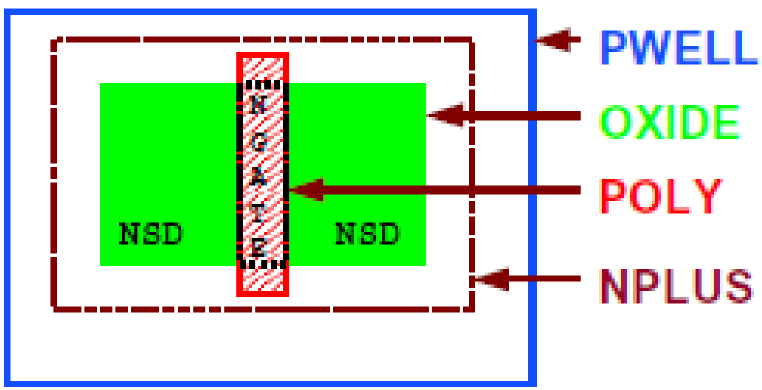
דוגמא: שכבות רכיבים

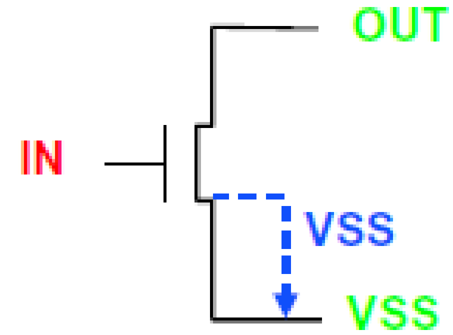
LAYER PWELL 1
LAYER OXIDE 2
LAYER POLY 4
LAYER NPLUS 5

ORIGINAL LAYERS

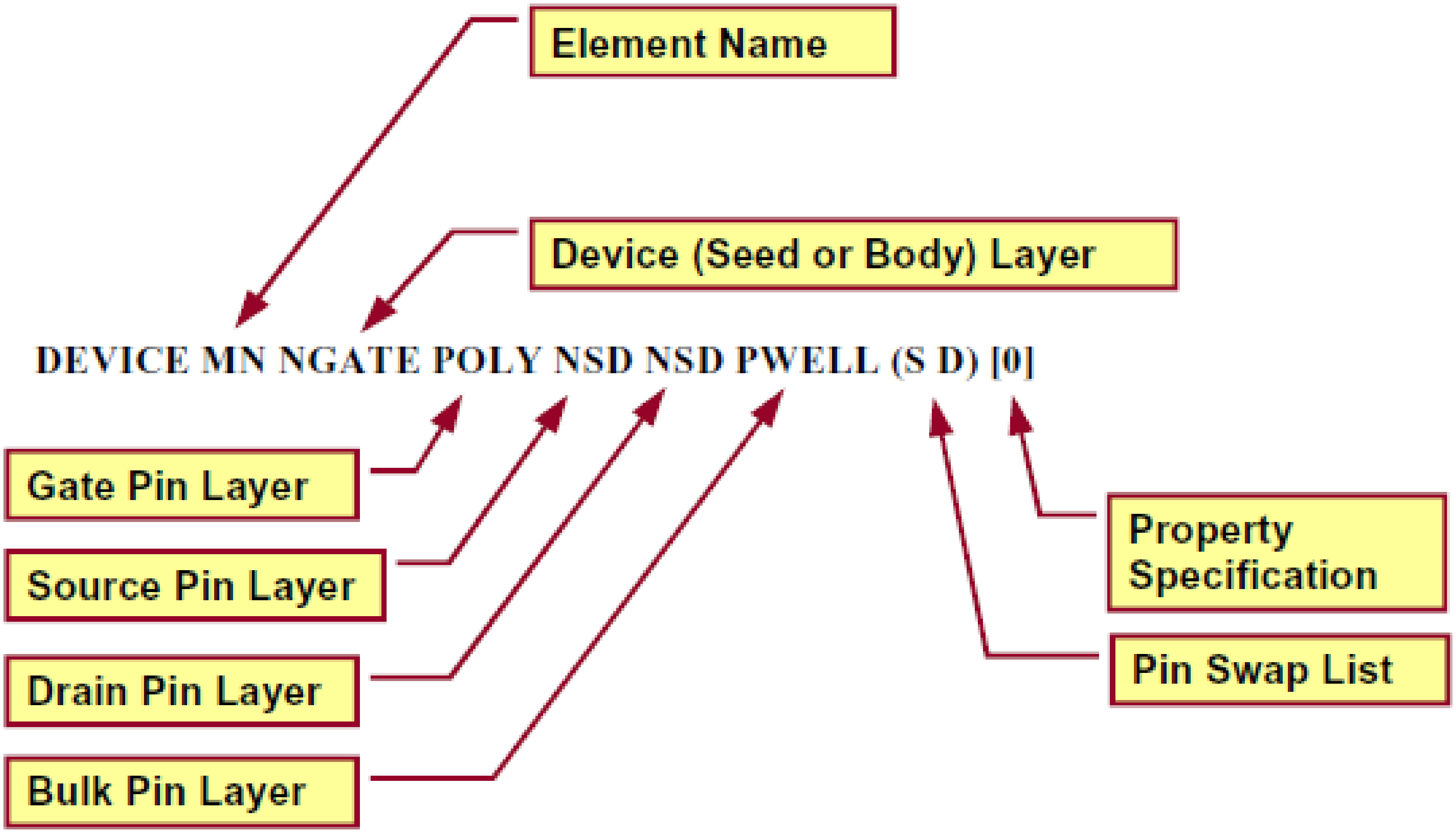
PAREA = OXIDE AND PWELL
NGATE = POLY AND PAREA
NOX = OXIDE AND NPLUS
NSD = NOX NOT NGATE

LAYER CONSTRUCTORS



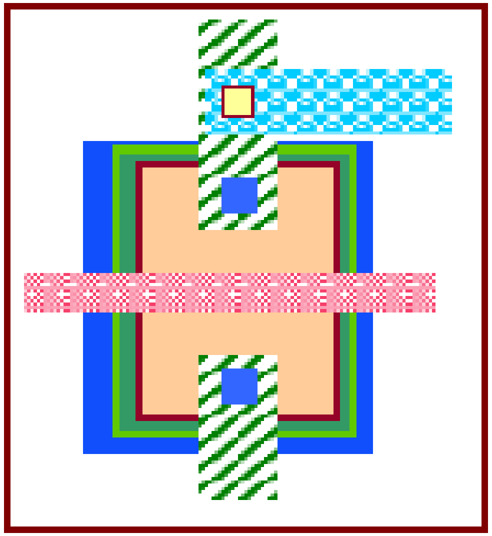


דוגמא: הצהרת רכיבים

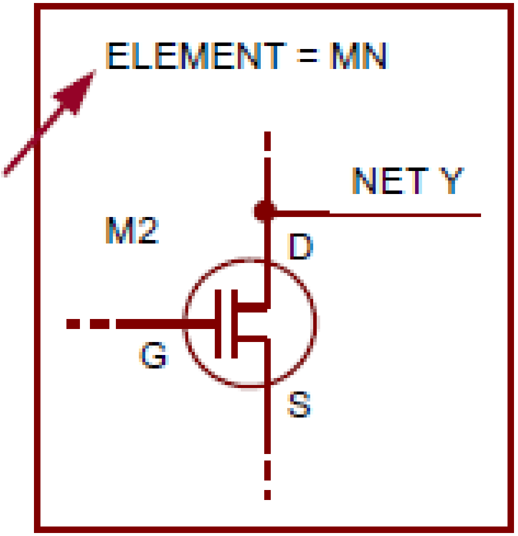


קשר בין layout ומקור הרכיבים

// Device Statement
DEVICE MN NGATE POLY NSD NSD PWELL [0]



1. LVS defines the MN element template based upon the Device statement.

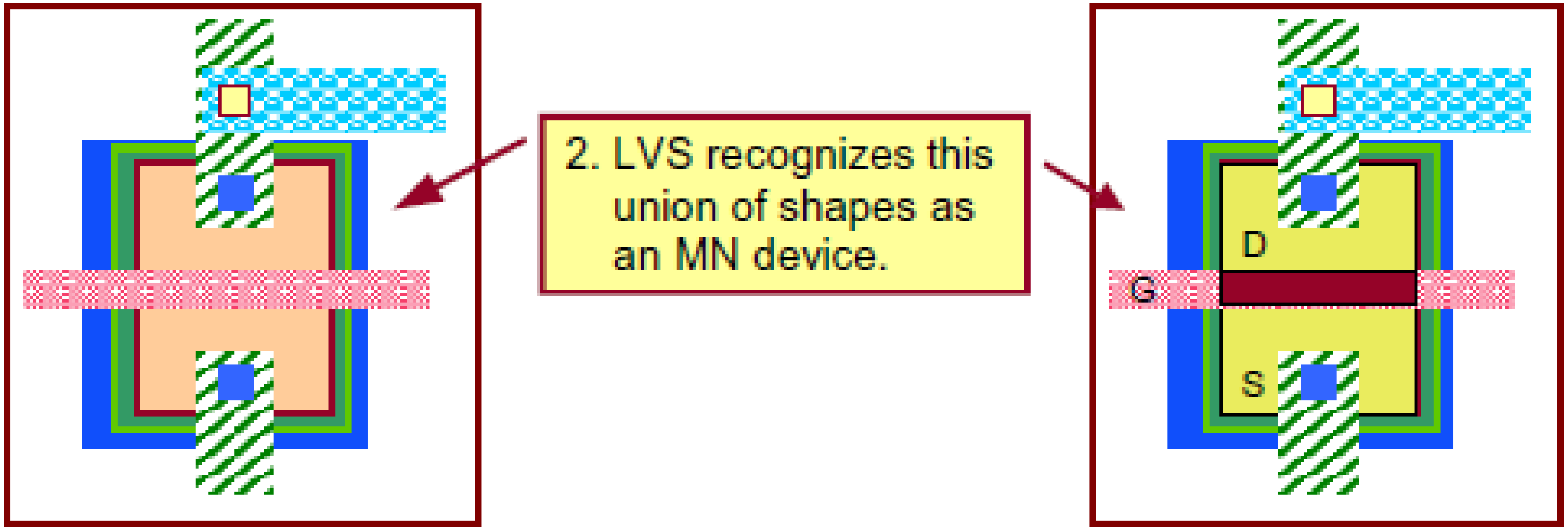


LAYER PWELL	1	
LAYER OXIDE	2	
LAYER POLY	4	
LAYER NPLUS	5	

PAREA	= OXIDE AND PWELL
NGATE	= POLY AND PAREA
NOX	= OXIDE AND NPLUS
NSD	= NOX NOT NGATE

קשר בין layout ומקור הרכיבים (המשך)

// Device Statement
DEVICE MN NGATE POLY NSD NSD PWELL [0]

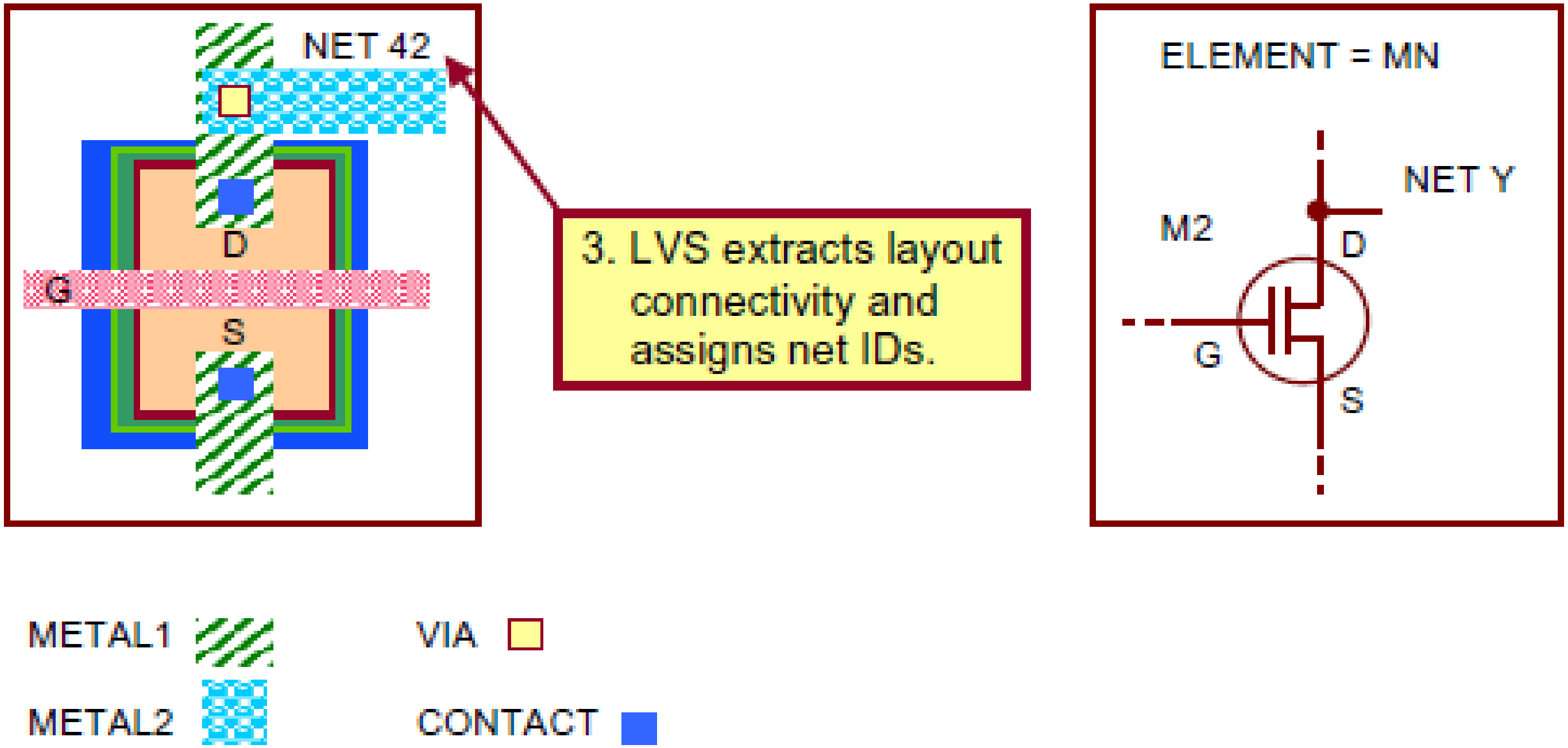


LAYER PWELL 1
LAYER OXIDE 2
LAYER POLY 4
LAYER NPLUS 5

PAREA = OXIDE AND PWELL
NGATE = POLY AND PAREA
NOX = OXIDE AND NPLUS
NSD = NOX NOT NGATE

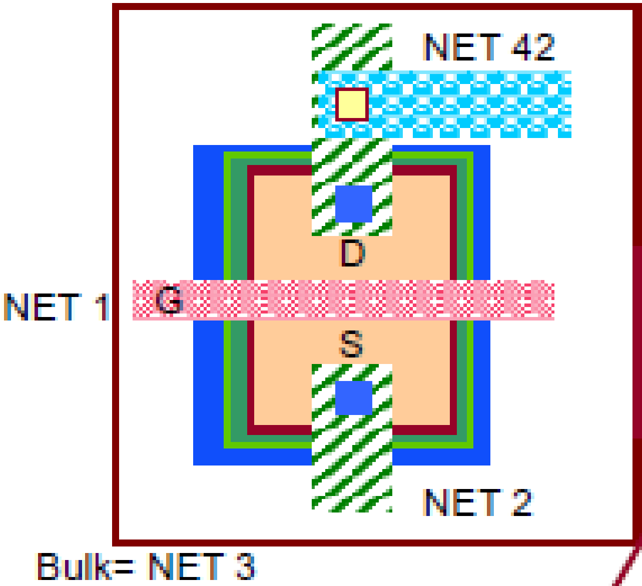
Seed Layer = NGATE
Gate = POLY
Source = NSD
Drain = NSD
Bulk = PWELL

קשר בין layout ומקור הרכיבים (המשך)

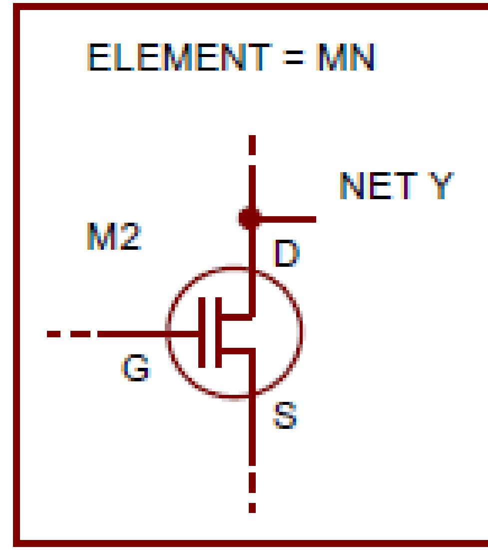



```
//Connection statements
CONNECT METAL1 METAL2 by VIA
CONNECT METAL1 NOX by CONTACT
```

קשר בין layout ומקור הרכיבים (המשך)



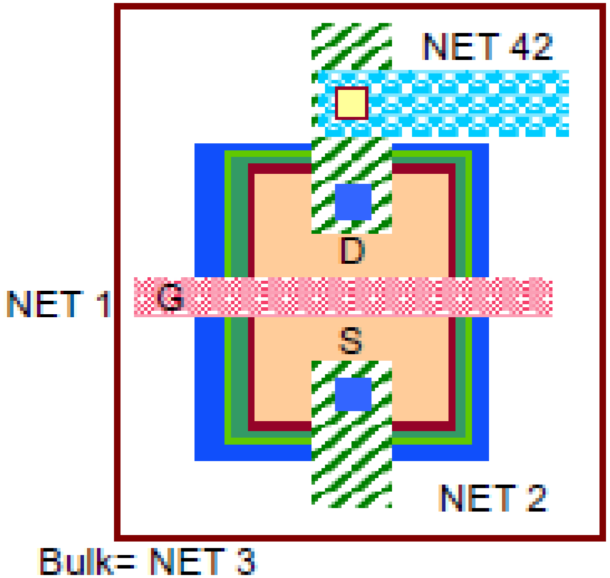
4. LVS associates device pins with net IDs to form the layout netlist.



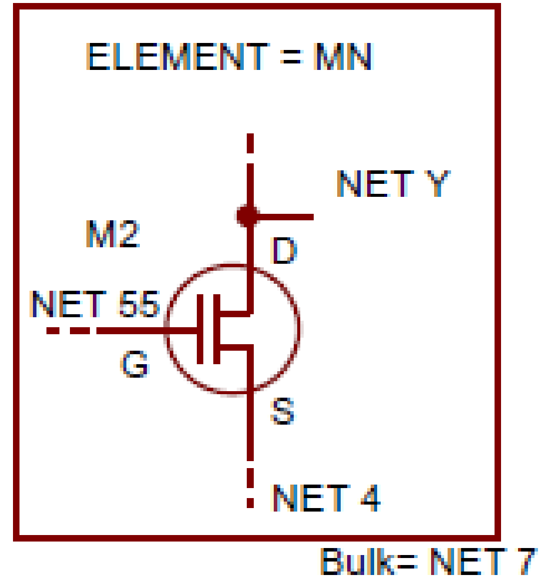
//Layout Netlist
M0 42 1 2 3 n L=1.25e-6 W=1.5e-5

NOTE: Nets 1 2 42 and 3 are arbitrary numbers

קשר בין layout ומקור הרכיבים (המשך)



5. LVS compares the layout netlist to the source netlist.



//Layout Netlist
M0 42 1 2 3 n L=1.25e-6 W=1.5e-5

//Source Netlist
M2 Y 55 4 7 n L=1.25e-6 W=1.5e-5

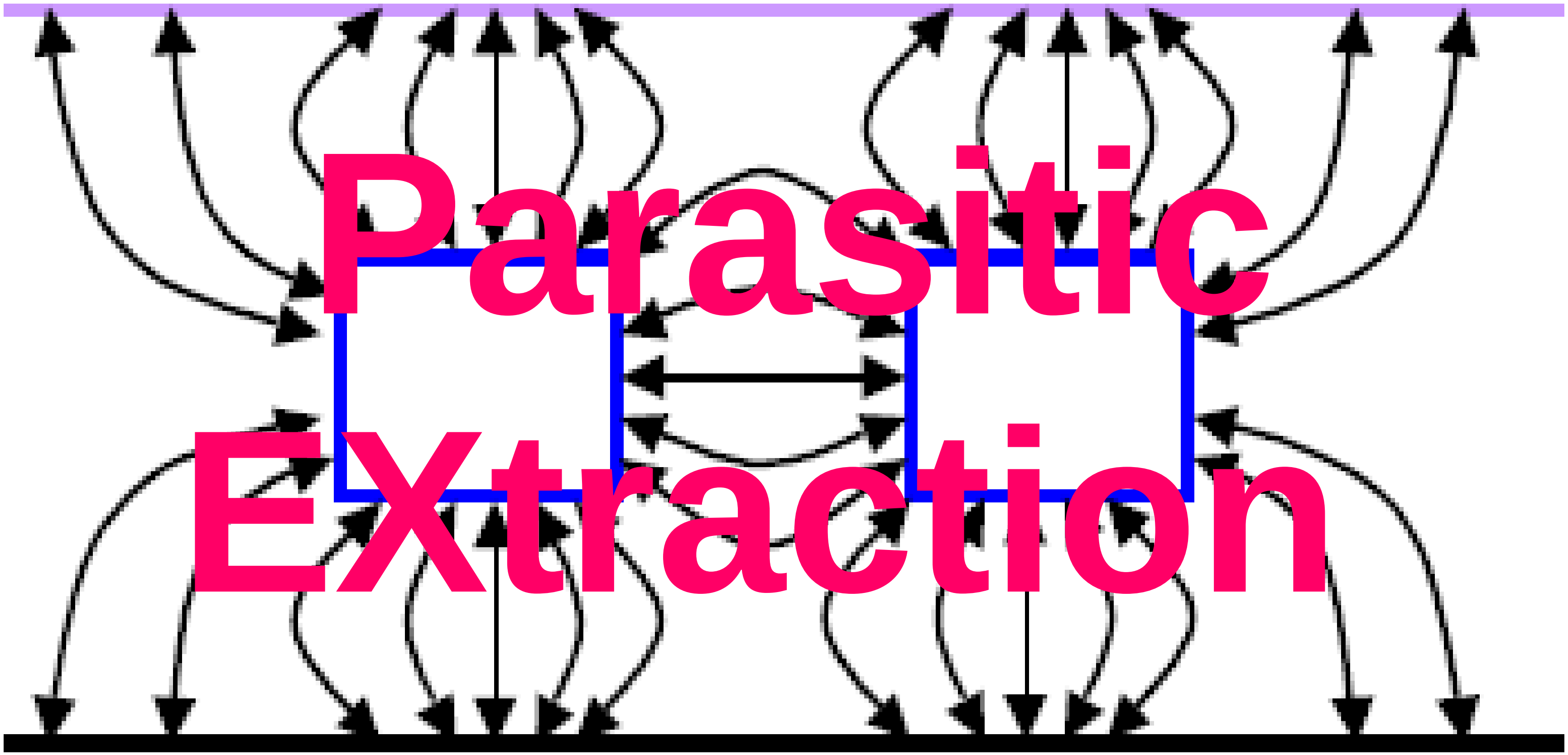
Device matches at this point, additional connectivity information is required for complete match (what other devices/instances share these nets)

דוגמת לגדרת רכיב

```
DEVICE MN(ND) bngate poly(G) tndiff(S)
tndiff(D) psub(B) <nthin>[
property W,L,AS,AD,PD,PS,NRD,NRS
W=(perimeter_coincide(bngate, tndiff ) +
perimeter_inside(bngate, tndiff)) / 2
L=area(bngate) / W
PI_S_OD = perimeter_inside(S,nthin)
IF(PI_S_OD > 0) {
AS = area(S) * W /PI_S_OD
PS = perimeter(S) * W /PI_S_OD
} ELSE { AS=0 PS=0 }
PI_D_OD = perimeter_inside(D,nthin)
IF(PI_D_OD > 0) {
AD = area(D) * W /PI_D_OD
PD = perimeter(D) * W
/PI_D_OD
} ELSE { AD=0 PD=0 }
#IFDEF ZERO_NRS_NRD
NRS = 0
NRD = 0
#ELSE
NRS = AS / W / W
NRD = AD / W / W
#ENDIF
]
```

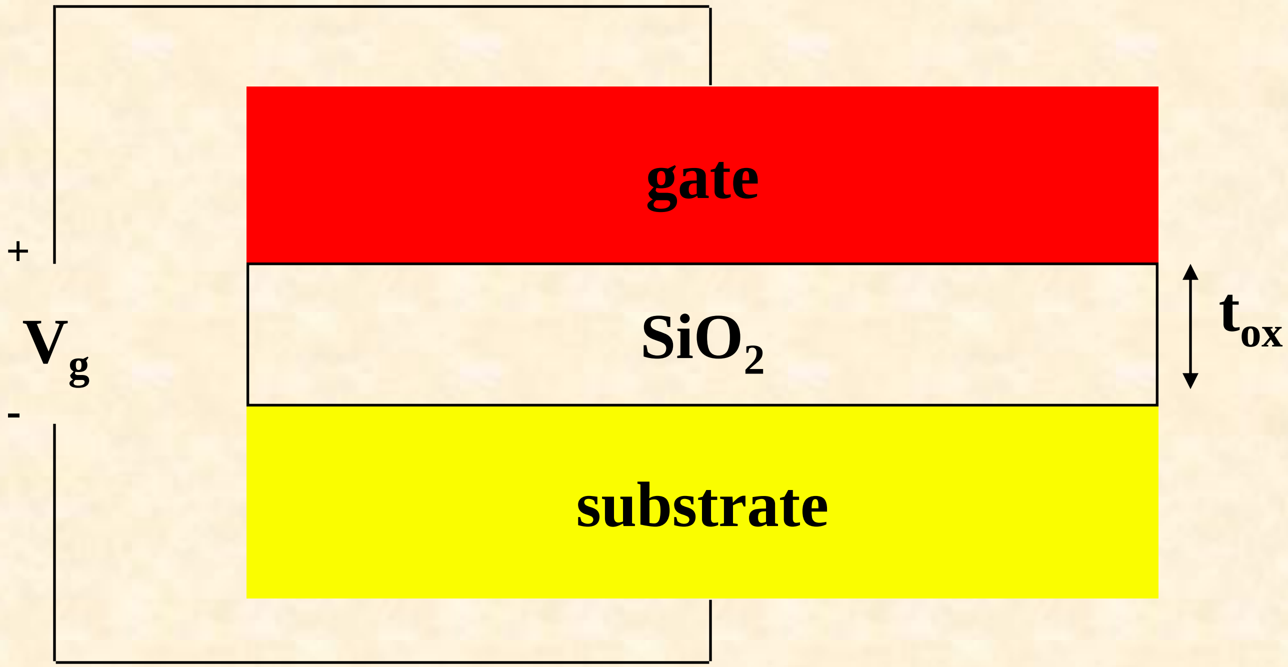
CalibreLVS.htm

Parasitic Extraction



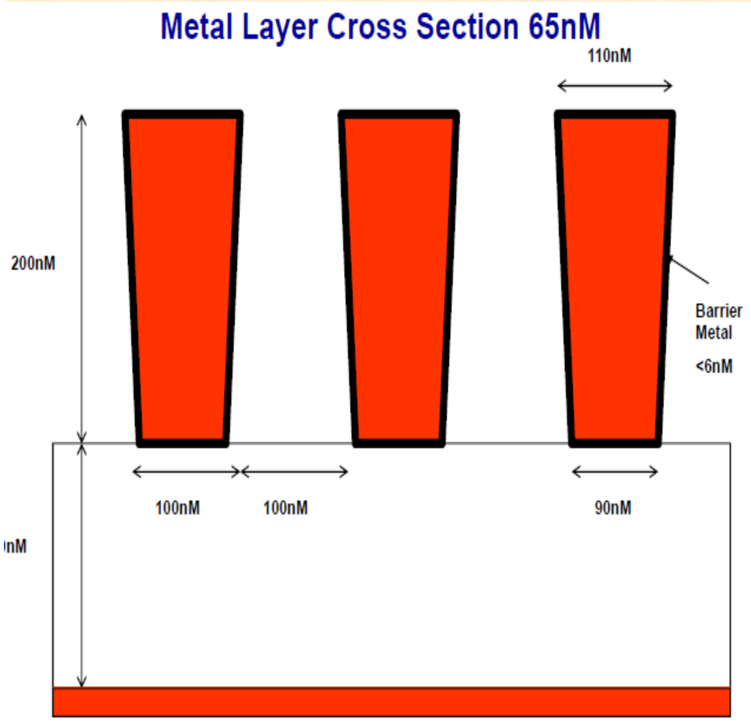
Parasitic Extraction

- הערכת קיבול בין מבנים ב layout –
- חישוב של התנגדות של החוטים



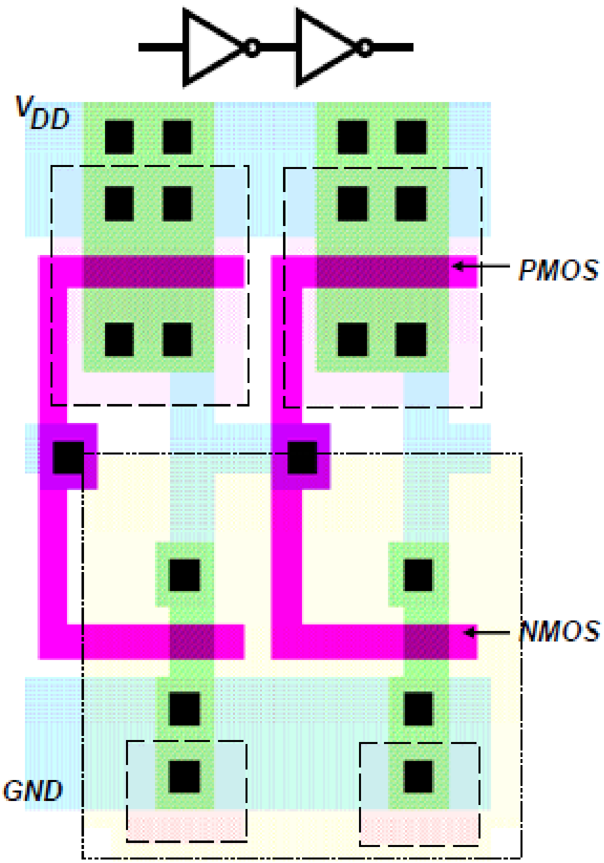
Interconnects -- החיבורים

- החיבורים הם חוטי סרט דקים שמחברים חשמלית שני רכיבים או יותר במעגל
- החיבורים יכולים להכניס מרכיבי קיבול, התנגדות, השראות לא רצויים.
- המרכיבים עלולים להשפיע לרעה
  – על הביצועים (כגון עיכוב)
  – על צריכת החשמל
  – על אמינות

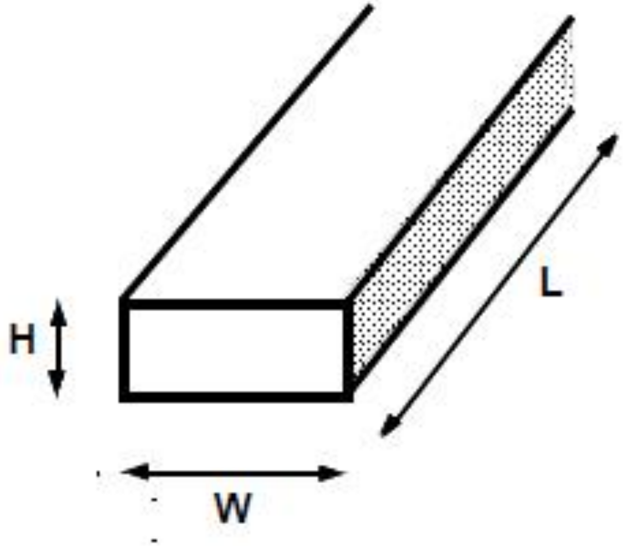


החיבורים

- הגדולים של טרנזיסטורים נופלים למטה ומספר חוטי מתכת גודל. לפי כך, ההשפעה של החיבורים הולכת וגודלת.
- קווי מתכת עבים רצים מעל שכבת תחמוצת (oxide) שמכסה את המצע
- הם תורמים להתנגדות ולקיבול



התנגדות חוטים



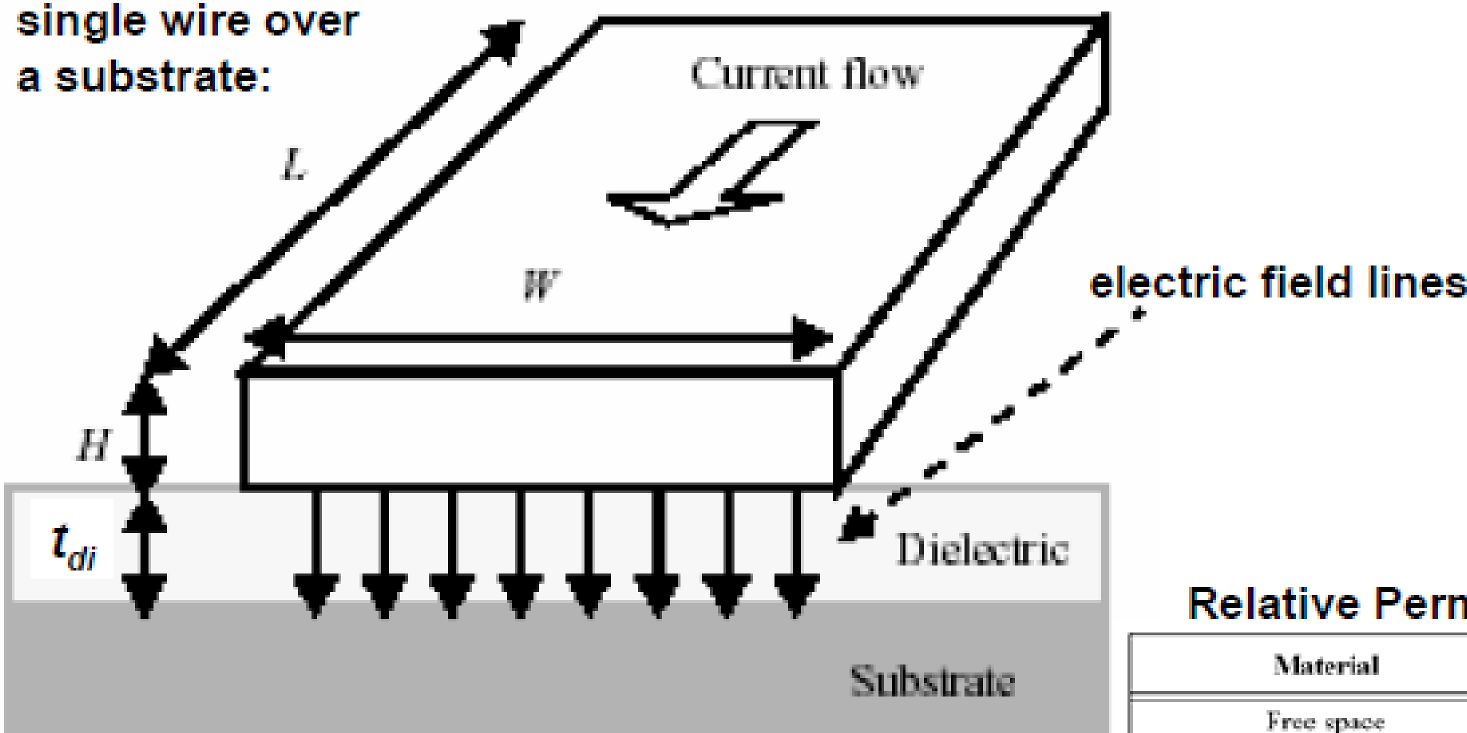
$$R_{wire} = \frac{\rho L}{H W} = R_s \frac{L}{W}$$

Material	ρ (Ω-m)
Silver (Ag)	1.6×10^{-8}
Copper (Cu)	1.7×10^{-8}
Gold (Au)	2.2×10^{-8}
Aluminum (Al)	2.7×10^{-8}
Tungsten (W)	5.5×10^{-8}

Material	Sheet Res. (Ω/□)
n, p well diffusion	1000 to 1500
n+, p+ diffusion	50 to 150
n+, p+ diffusion with silicide	3 to 5
polysilicon	150 to 200
polysilicon with silicide	4 to 5
Aluminum	0.05 to 0.1

קיבול חוטים

single wire over
a substrate:

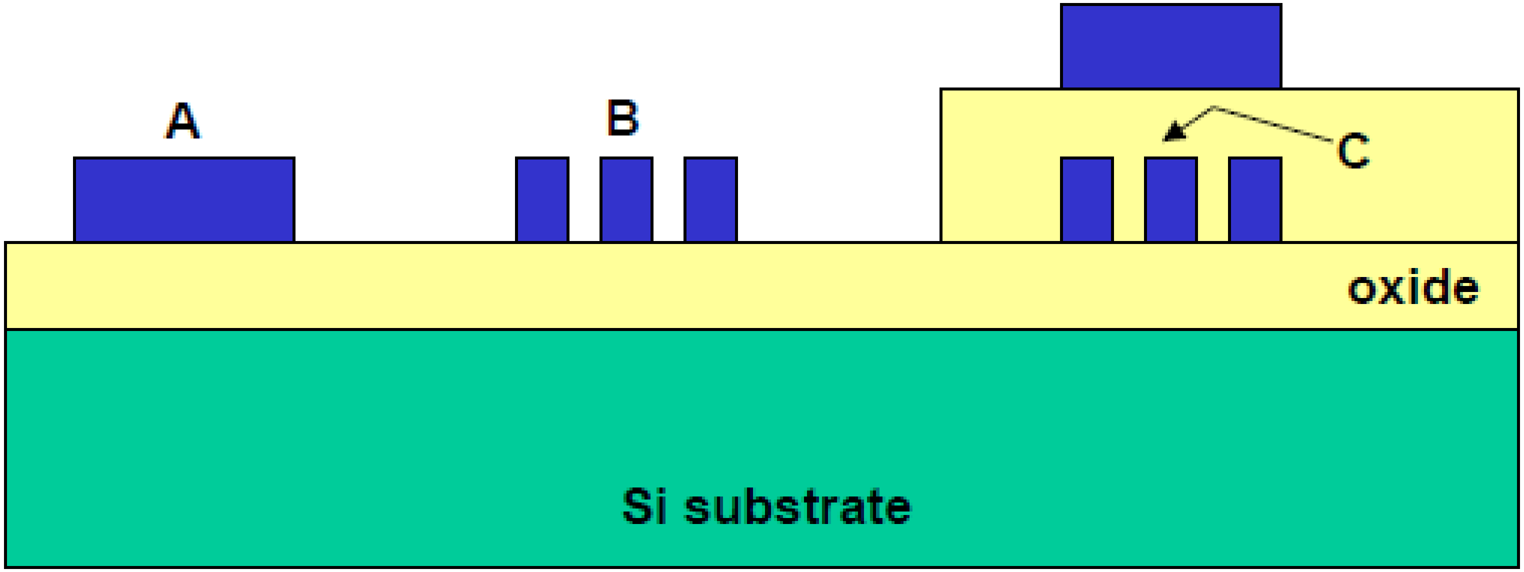


$$C_{pp} = \frac{\varepsilon_{di}}{t_{di}} WL$$

Relative Permittivities

Material	ε_r
Free space	1
Aerogels	~1.5
Polyimides (organic)	3-4
Silicon dioxide	3.9
Glass-epoxy (PC board)	5
Silicon Nitride (Si_3N_4)	7.5
Alumina (package)	9.5
Silicon	11.7

קיבול של חיבור חוט ל-חוט



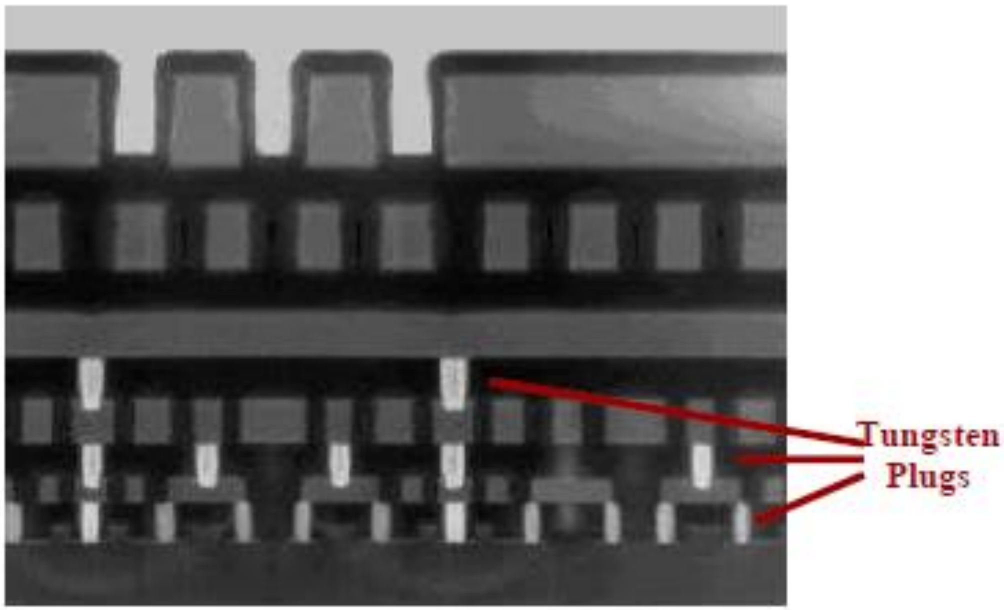
Wire A simply has capacitance (C_{pp} + C_{fringe}) to substrate

Wire B has additional sidewall capacitance to neighboring wires

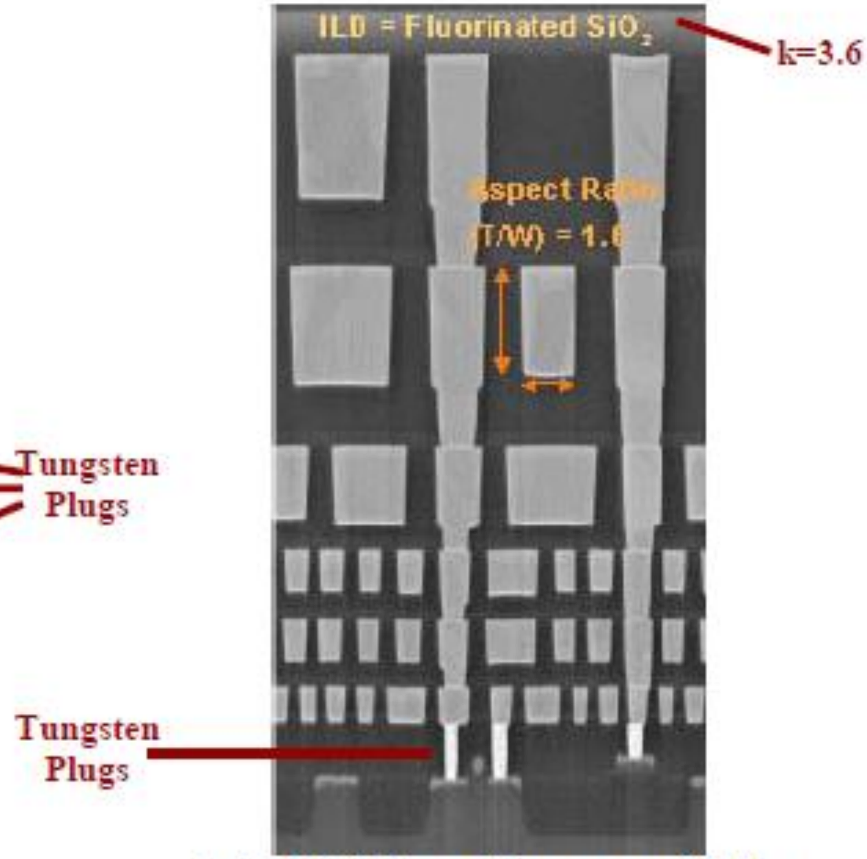
Wire C has additional capacitance to the wire above it

דוגמאות חיבור: INTEL

Advanced processes: narrow linewidths, taller wires, close spacing → relatively large inter-wire capacitances



Intel 0.25µm Process (Al)
5 Layers - Tungsten Vias
Source: *Intel Technical Journal* 3Q98



Intel 0.13µm Process (Cu)
Source: *Intel Technical Journal* 2Q02

השפעת קיבול בין-חוטי

הקיבול בין החוטים הצפופים מוביל שתי השפעות עיקריות:

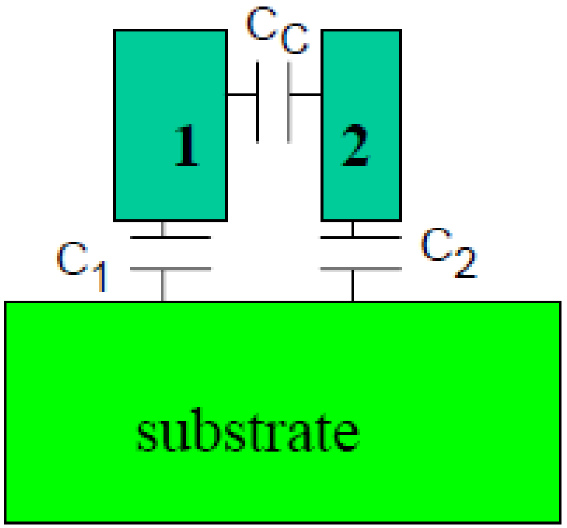
1. הקיבול המוגבר מביא להפסד מהירות
2. העברה של אותות לא רצוייה ממקום אחד למקום אחר דרך הקיבול שנקראת "Crosstalk"

השפעת קיבול בין-חוטי

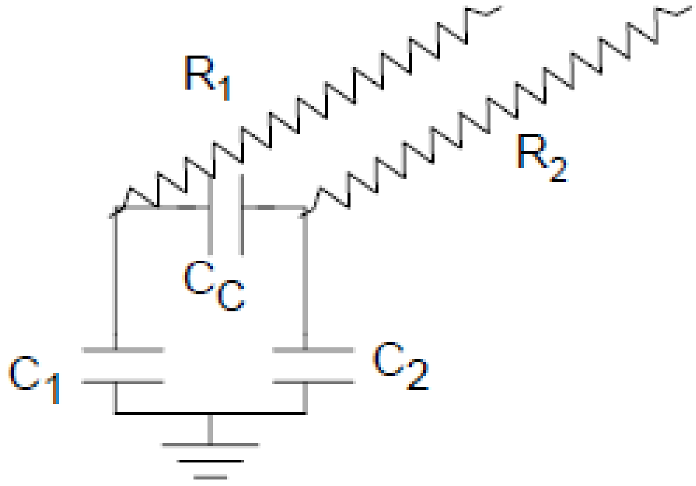
There are three capacitances as illustrated

Wire 1 has resistance R_1

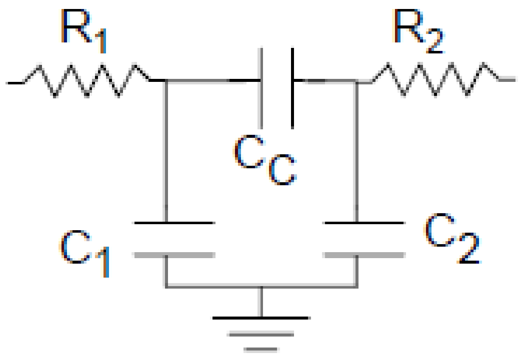
Wire 2 has resistance R_2



Using a simple lumped model for each wire we have three capacitances and two resistances



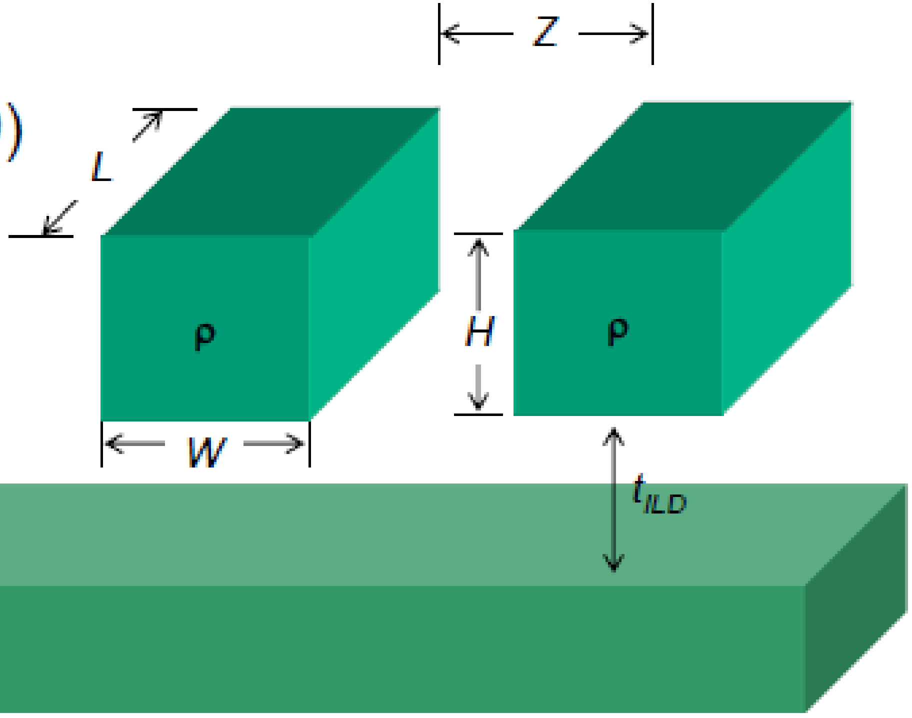
Which, when redrawn in a plane, has a "π " shape



דירוג חיבורים

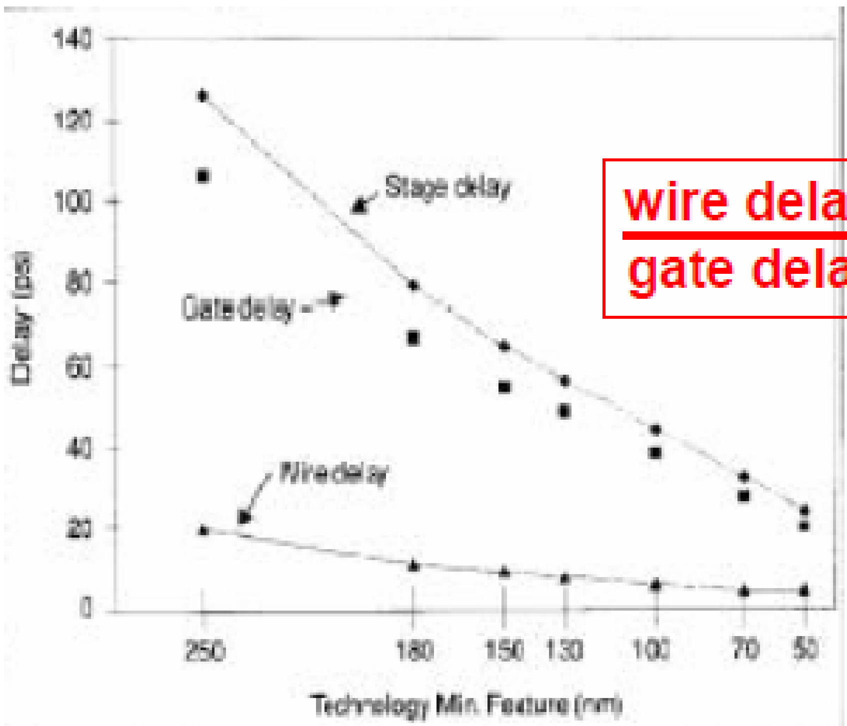
Relevant parameters:

- wire width W
- wire length L
- wire thickness H
- **wire resistivity** ρ
- wire-to-wire spacing Z
- inter-level dielectric (ILD)
  - thickness t_{ILD}

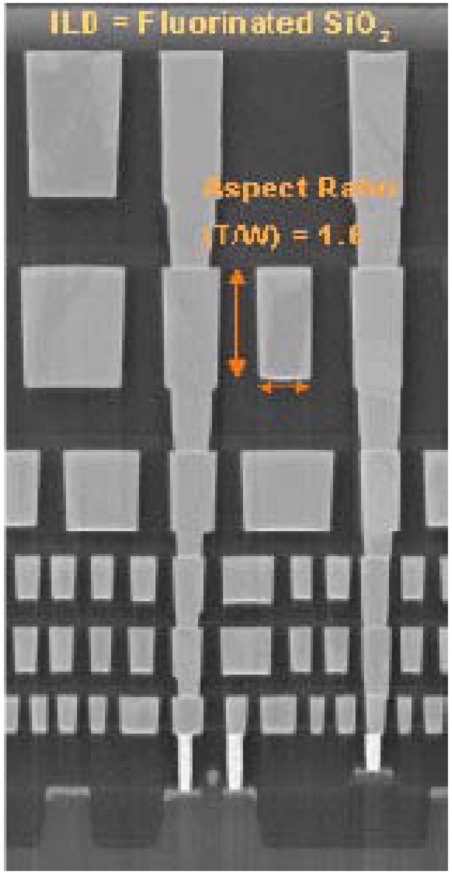


מגמות טכנולוגיות של חיבורים

- **Reduce the inter-layer dielectric permittivity**
  - "low-k" dielectrics ($\varepsilon_r \cong 2$)
- **Use more layers of wiring**
  - average wire length is reduced
  - chip area is reduced



$\frac{\text{wire delay}}{\text{gate delay}}$ increases



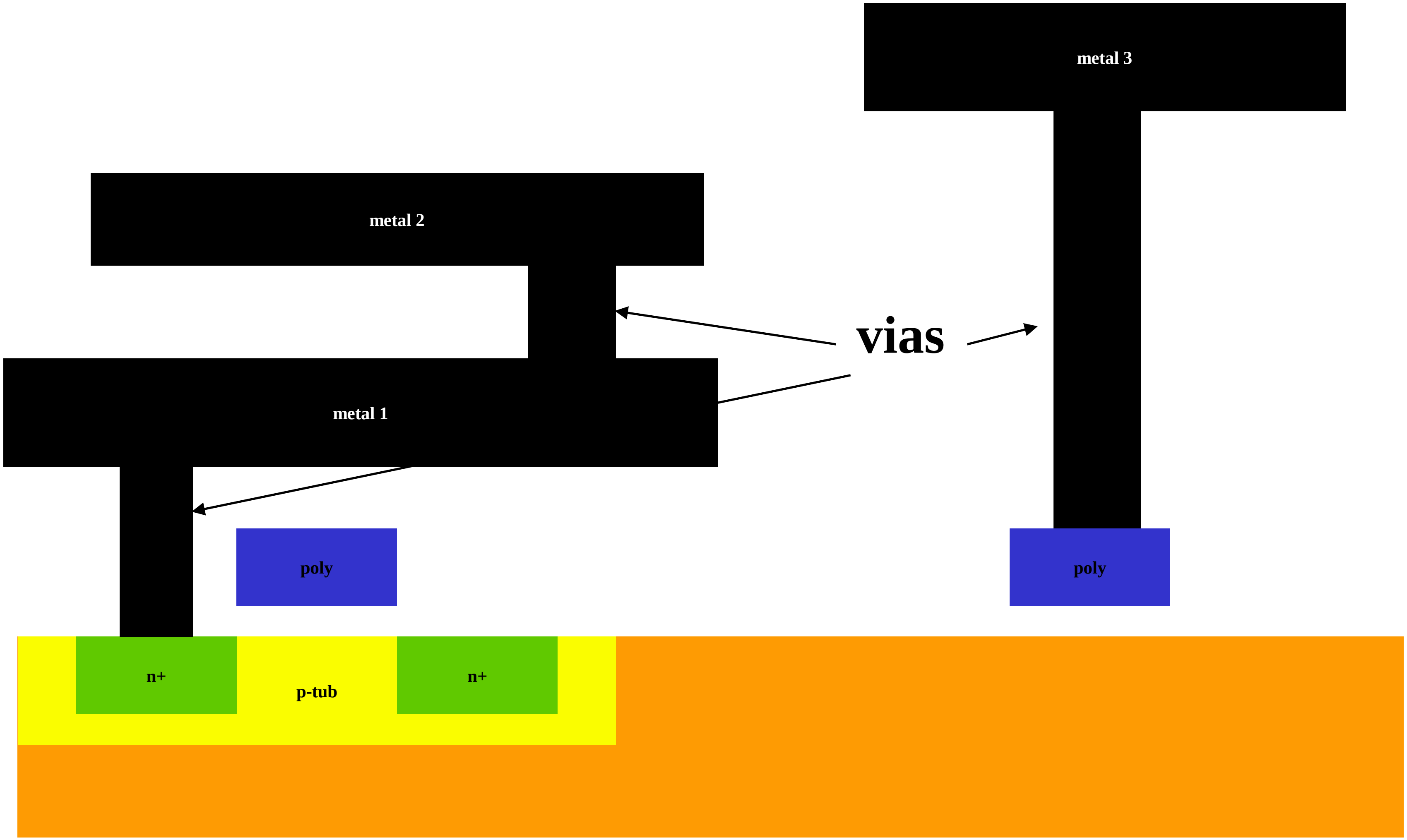
Intel 0.13µm Process (Cu)

Source: ***Intel Technical Journal*** **2Q02**

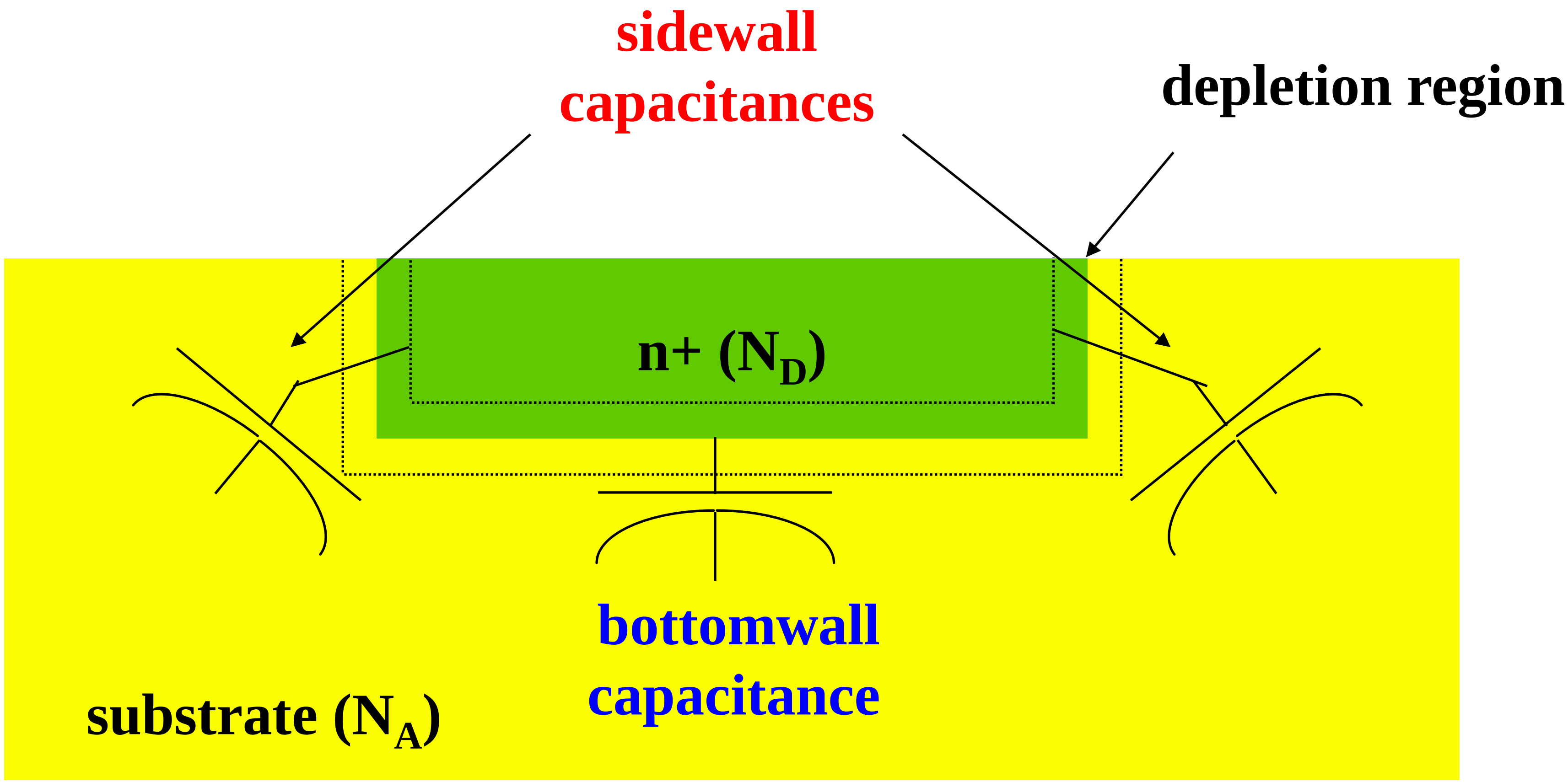
Contacts &Vias

- מגעים דומים לקווי מתכת
- ההתנגדות שלהם עולה כאשר הצפיפות עולה
- כאשר הגיאומטריה הולכת וקוטנת, המרווחים בין המגעים צריכים להיות מוגדלים
- זה יגרום לפחות מגעים בטרנזיסטור
- כאשר הזרם דרך הטרנזיסטור גודל, ההתנגדות של המגעים הופך להיות בעיה גדולה מאוד
- הצעד כדי לשפר את ההתנגדות של המגעים הוא להשתמש בנחושת

מגעים

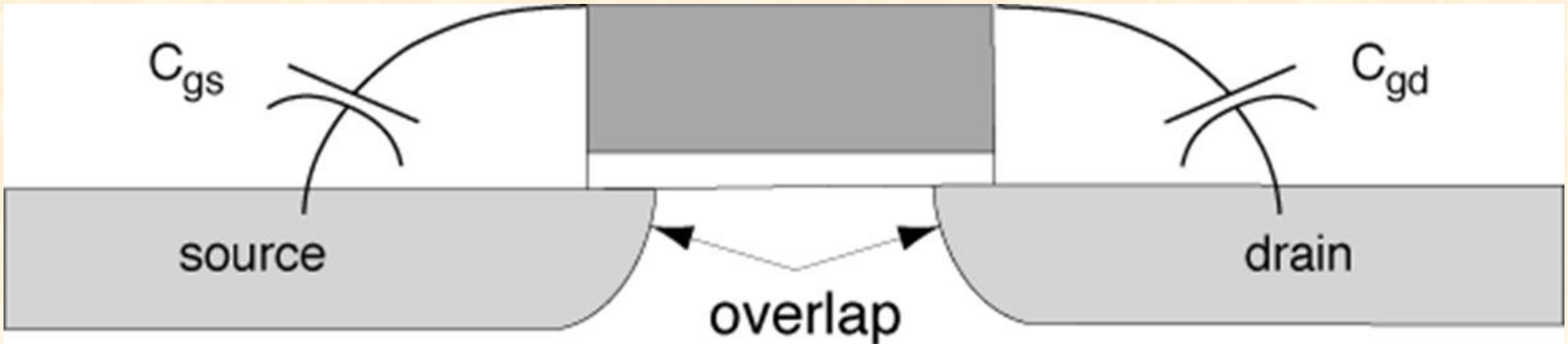


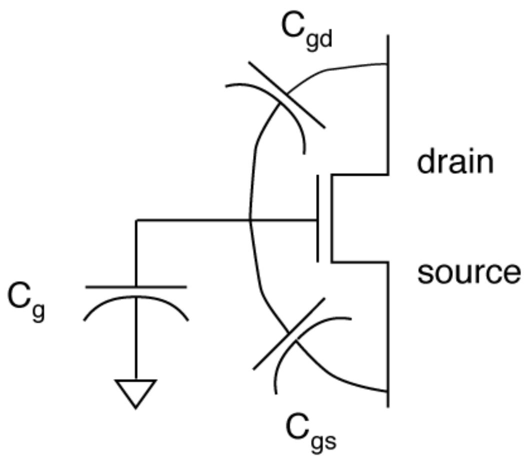
קיבול דיפוזיה

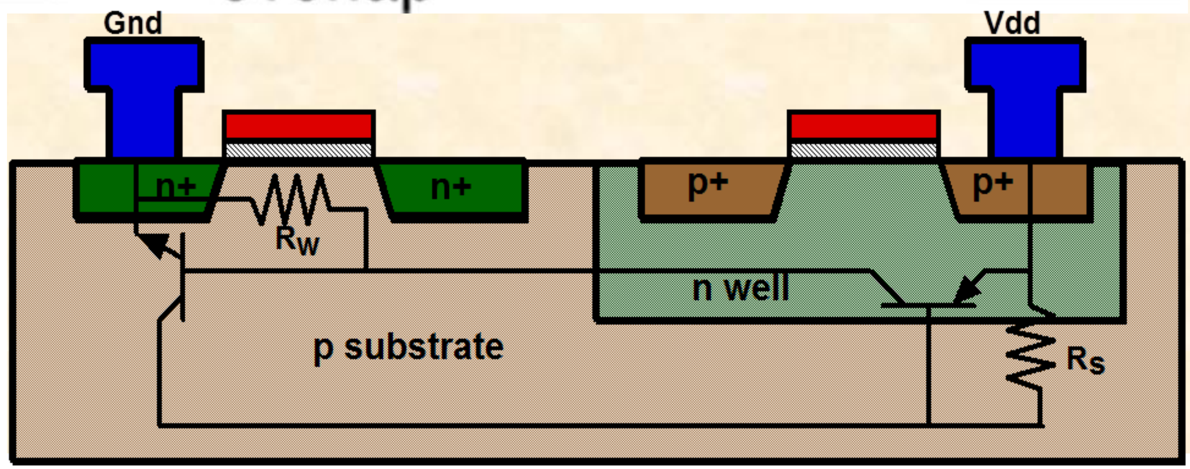


קיבול והתנגדות בטרנזיסטור

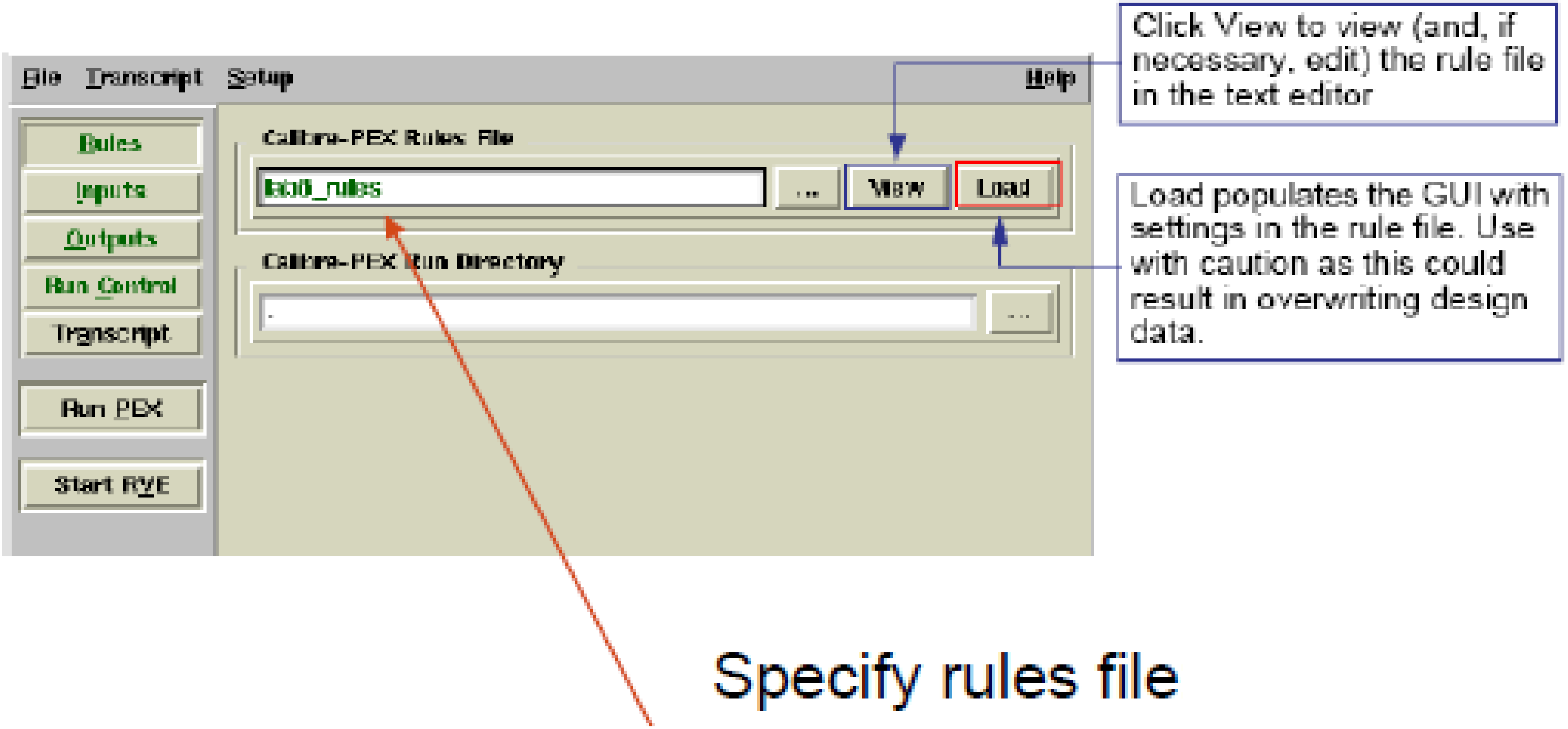
- קיבול בין השער למצע, גם בין השער למקור /ניקוז
- קיבול בין מקור לניקוז





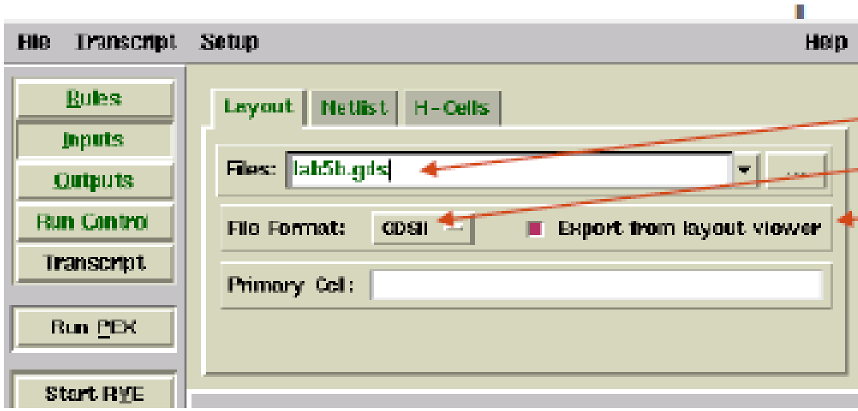


PEX runset



File Transcript Setup
Help
Rules
Inputs
Outputs
Run Control
Transcript
Run PEX
Start RVE
Calibre-PEX Rules File
lab8_rules
View
Load
Calibre-PEX Run Directory
Click View to view (and, if necessary, edit) the rule file in the text editor
Load populates the GUI with settings in the rule file. Use with caution as this could result in overwriting design data.
Specify rules file

קלט ל-PEX

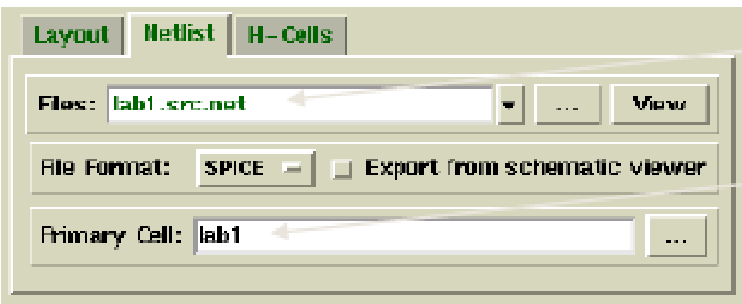


Name of layout file (count4.gds)

GDSII file format

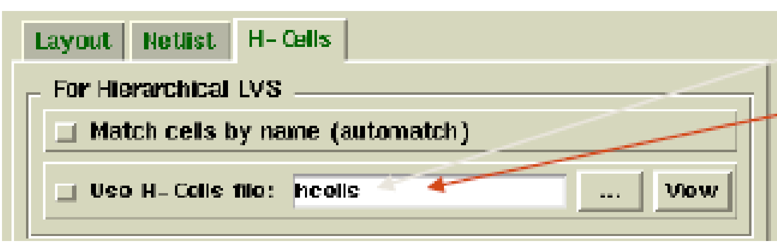
Check to generate new layout file

Name of top cell (count4)



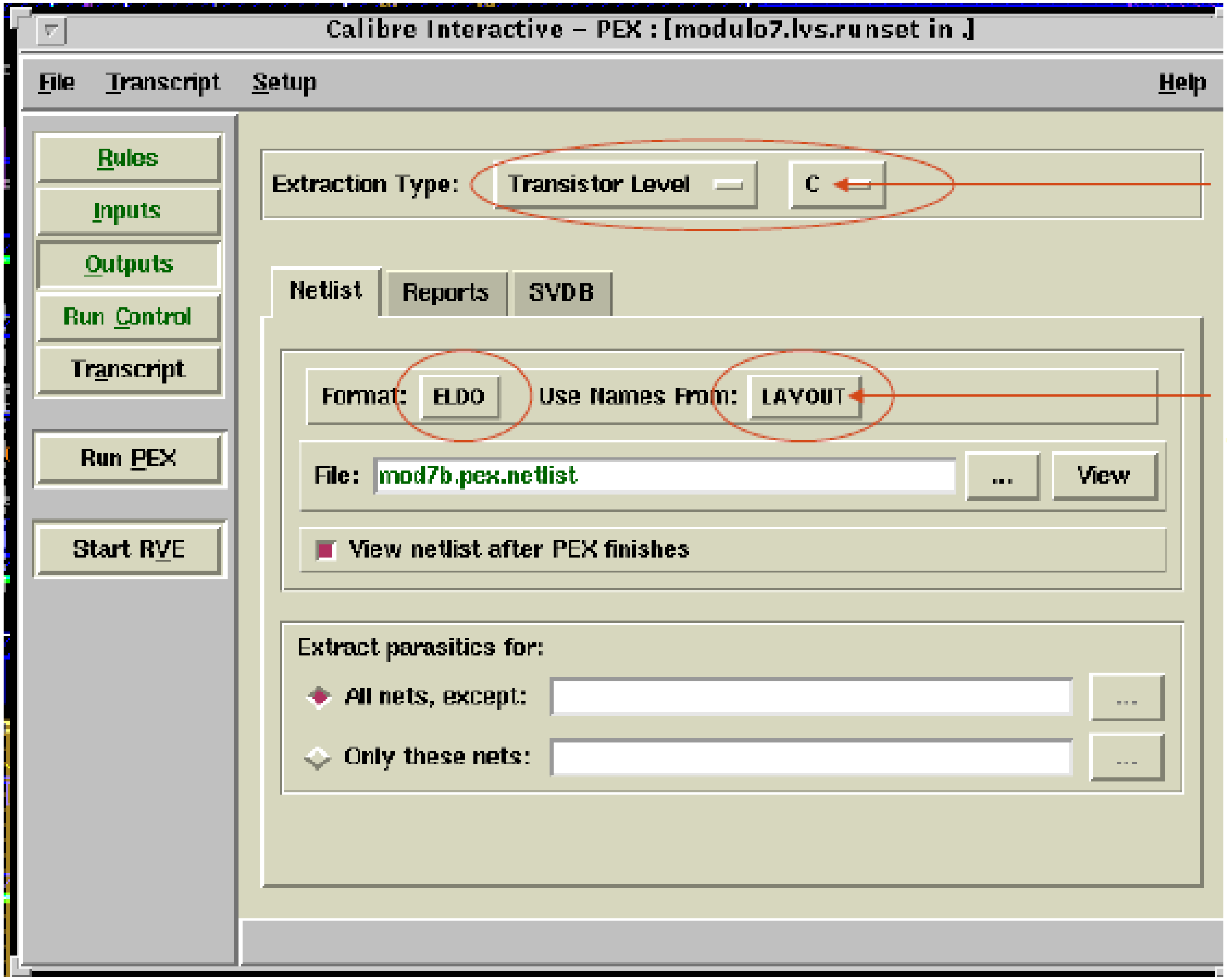
Source(SPICE) netlist

Top-level cell name in SPICE netlist



Hierarchical cells file:

פלט של PEX



Lumped capacitance

Use net names from LAYOUT